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A Diagnostic Evaluation Framework for Time-Series–Text Alignment Models

Master's Thesis

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Abstract

Time-series–text alignment models match numerical time series with natural-language descriptions and are evaluated almost exclusively by how often they return the correct match. Such a score does not show which information in the input the model uses. We present a controlled diagnostic framework that addresses this question by changing one part of the input at a time and comparing each model only with itself on the unaltered input. D1 swaps a single component of a description, D2 destroys the temporal order of the series while keeping every value, and D3 replaces the series with inputs that retain progressively less statistical information. Each diagnostic includes validity controls: a metric audit against an untrained model, capability gating, human-certified test items, equivalence tests, and a text-only negative control. We apply the framework to a reimplementation of CLaSP, the released TRACE retriever and ChatTS-14B, with `text-embedding-3-large` as negative control, on SUSHI, TRUCE and TRACE's NOAA weather data. The metric audit shows that, in one configuration, the judge-based mAP@10 used to evaluate CLaSP gives trained and untrained models similar scores, so we use strict retrieval throughout. Every model with interpretable results shows a significant drop on at least one description component, and the component on which it does worst differs by model: fluctuation type for CLaSP, trend family for ChatTS and temperature-trend direction for TRACE. Full shuffling removes 66–98% of retrieval MRR on order-dependent descriptions and brings ChatTS to chance. The small residual that survives has different sources: the shape of the value distribution (CLaSP on SUSHI), distribution shape and/or cross-channel co-occurrence (TRACE), and a series-length cue created by the merged candidate pool (CLaSP on TRUCE). The negative control yields a positive difference-in-differences without any matching ability, which shows that an order-sensitivity statistic can mislead on its own. Taken together, a high matching score does not explain itself. Controlled diagnostics with built-in validity checks can show what lies behind it.

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Chapter 1

Introduction

1.1 Background and problem

Models that align numerical time series with natural language are being developed for search and question answering over sensor, financial and environmental data. Retrieval models such as CLaSP (Ito et al., 2025) and TRACE (Chen et al., 2026), which retrieve series using natural-language queries, embed a series and a description separately and rank candidates by similarity. Time-series language models such as ChatTS (Xie et al., 2024) read the series and the text jointly. In this thesis, *time-series–text alignment* means whether the meaning of a description matches the properties of a series: its trend, a local event, or the order in which events occur. Figure 1.1 shows an example.

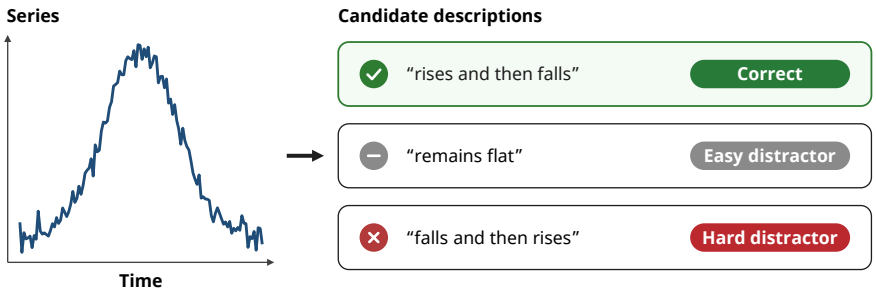


Figure 1.1: Time-series–text alignment. A model must decide which description matches the series. Rejecting “remains flat” requires only detecting that the series moves up and down; rejecting “falls and then rises” requires using the order of the two movements.

These systems are evaluated almost exclusively by aggregate correctness (Recall@k, mean reciprocal rank (MRR), or multiple-choice accuracy). A high score, however, does not say what information the model used to obtain it. In Figure 1.1, a model can reject “remains flat” by detecting that the series contains both an increase and a decrease, without using order. This does not show that it can reject “falls and then rises”, which differs only in the order of the two movements. Decision rules that succeed under the standard evaluation but fail when its conditions change are called *shortcuts* (Geirhos et al., 2020). Whether a rule is a shortcut depends on what the description claims: summary statistics suffice to verify a claim about the mean, not one about the order of events. When the alternatives in an evaluation differ in many properties at once, a correct match cannot be attributed to any single one.

1.2 Research gap and questions

Controlled input changes that alter one property while holding the rest fixed are the standard tool for this question, and each has been applied in isolation in a neighbouring setting. Component swaps, which change one part of a description (e.g., "a red cube" to "a blue cube"), exposed that vision–language retrievers with strong scores often fail to bind attributes and relations (Yuksekgonul et al., 2022); shuffling the order of the values in a time series showed that LLM-based forecasters are no more order-sensitive than their LLM-free ablations (Tan et al., 2024); and ablations of numbers in the prompt showed that a time-series LLM can lean heavily on statistics written as text (Yin et al., 2026; Xie et al., 2024). No study applies all three kinds of control to the *same* time-series–text alignment models. It therefore remains unknown whether these models are sensitive to individual parts of a description, whether they use temporal order, and what supports the performance that survives when order is removed.

This thesis addresses the main question: ***what information do time-series–text alignment models rely on when they match a series with a description?*** We develop three diagnostics (Figure 1.2). Each diagnostic changes one part of the input and compares a model only with itself on the unchanged input:

RQ1 (D1, *component sensitivity*): can a model reject a description that differs from the correct one in exactly one component, such as trend direction or fluctuation type?

RQ2 (D2, *temporal-order sensitivity*): does matching survive when the order of the values is destroyed but every value is kept, and is any loss specific to descriptions that make claims about order?

RQ3 (D3, *statistical-information sufficiency*): if some performance survives shuffling, what information (exact values, the value distribution, mean and spread, or length) supports it?

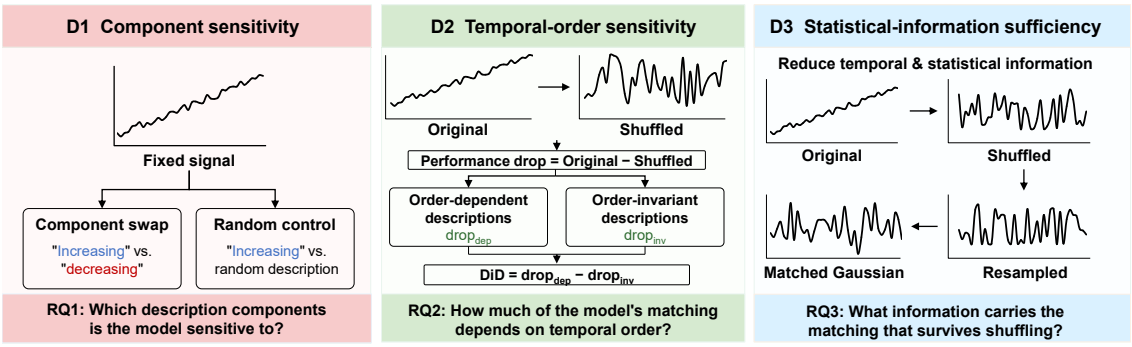


Figure 1.2: **The three diagnostics.** D1 keeps the signal fixed and asks a model to choose between the correct description and a copy with one component swapped, against a matched random-description control. D2 perturbs the order of the series and contrasts descriptions that make claims about order with those that do not. D3 replaces the series with inputs that keep progressively less information, to locate what supports the performance that survives shuffling. All signals are synthetic.

1.3 Contributions

This thesis makes three key contributions.

(i) We propose a model-agnostic framework with three diagnostics for time-series–text alignment: D1 component sensitivity, D2 temporal-order sensitivity and D3 statistical-information sufficiency. They are developed with explicit interpretation rules and work

both for retrieval models, which rank candidates by a similarity score, and for time-series language models such as ChatTS, which are asked here to choose between two descriptions.

(ii) We build five validity controls into the framework that determine when a result can be interpreted: a metric audit against an untrained model, capability gating on each model's own reference performance, human certification of test items, equivalence tests for claims of "no change", and a text-only negative control that has no time-series encoder.

(iii) We report how three alignment models behave on three datasets: which description components each model handles poorly, how much of its performance requires temporal order, and what information supports the performance that survives shuffling.

1.4 Thesis outline

Chapter 2 reviews the background, the alignment models studied here, existing evaluation practice and the controlled diagnostics this thesis builds on.
Chapter 3 defines the three diagnostics, the validity controls and the verdict labels, and describes the models, datasets, item construction and statistical procedures.
Chapter 4 first checks the evaluation metric and each model's reference capability, and then answers **RQ1**–**RQ3** in turn.
Chapter 5 brings the diagnostics together, derives recommendations for evaluating alignment models, states the limitations, and concludes.
The appendices document the CLaSP reimplementation, item construction and human validation, the construction of every condition, all per-run results and tests, the predictions recorded in advance, and the code and model versions.

Chapter 2

Background and Related Work

This chapter reviews the technical background. It first explains how time-series representations have been linked to language. It then introduces the alignment models studied in this thesis and how they are usually evaluated, and finally reviews the controlled tests from related fields that this thesis builds on.

2.1 From time-series representations to language

Deep networks learn representations of the input directly from data, without relying on hand-specified features (LeCun et al., 2015; Bengio et al., 2013). For time series, this has led to convolutional, recurrent (Hochreiter and Schmidhuber, 1997) and, more recently, attention-based models (Vaswani et al., 2017; Zhou et al., 2021; Nie et al., 2022) for classification and forecasting (Ismail Fawaz et al., 2019; Lim and Zohren, 2021). Other work learns general representations that can be reused across tasks, either through contrastive training (Yue et al., 2022) or by converting values into tokens for a language-model architecture (Ansari et al., 2024). Pretrained language models have also been adapted as forecasters (Zhou et al., 2023; Jin et al., 2024). Where a language model or its architecture is used in these approaches, it is used to predict values, not to relate a series to a description of it.

Text also plays different roles in time-series data. It can provide context that helps prediction, such as news accompanying a financial series (Liu et al., 2024), or it can describe the observed signal itself, as in time-series captions (Jhamtani and Berg-Kirkpatrick, 2021; Dohi et al., 2025). Only captions make claims about the series that can be checked against it, and these are the focus of this thesis.

2.2 Time-series–text alignment models

In multimodal learning, alignment refers to relating elements of one modality to elements of another (Baltrušaitis et al., 2018). In vision–language learning, CLIP (Radford et al., 2021) addresses it with contrastive training: an image encoder and a text encoder are trained so that matching image–text pairs receive higher similarity scores than mismatched pairs. CLaSP (Ito et al., 2025) applies the contrastive approach of CLIP (Radford et al., 2021) to time series: it trains a time-series encoder and a T5 text encoder (Raffel et al., 2020) on SUSHI (Kawaguchi et al., 2024) and TRUCE (Jhamtani and Berg-Kirkpatrick, 2021). TRACE (Chen et al., 2026) also uses contrastive training, but for multivariate series paired with longer descriptions made of several text fields. It is also trained with hard negatives, wrong candidates chosen to be similar to the correct one, which are intended to make the model better at telling similar candidates apart. ChatTS (Xie et al., 2024) takes a different

approach: it passes embeddings of short segments of the series into a large language model together with the prompt, and answers questions in text.

These models can therefore be queried in two ways: through a similarity score that ranks candidates, or through a choice between written options. The diagnostics in this thesis are designed to work with both.

2.3 Evaluating time-series understanding

Retrieval models are usually evaluated with Recall@k, which checks whether the correct series appears among the top k results, and mean reciprocal rank (MRR), which gives more credit the higher the correct series is ranked. CLaSP is instead evaluated with a judge-based mAP@10: a retrieved series counts as relevant if a sentence encoder, such as Sentence-BERT (Reimers and Gurevych, 2019) rates its caption as similar enough to the query (Ito et al., 2025). Whether this judge can separate good retrieval from poor retrieval is examined in Section 4.1.

Recent benchmarks have broadened the range of abilities that are tested. These include recognising features such as trend, seasonality and anomalies (Fons et al., 2024), recognising and generating descriptions (Sen et al., 2025), structural awareness, temporal reasoning and cross-modal alignment (Yin et al., 2026), and retrieving events from long series (Zumarraga et al., 2026). These benchmarks show where models succeed or fail, but they are not designed to show *which information in the input a success depends on*. The diagnostics in this thesis address that question and can be applied to benchmark items that pair a series with its correct description.

2.4 Controlled diagnostics and shortcuts

Shortcut learning (Geirhos et al., 2020) motivates controlled tests that change one property of the input while keeping the rest fixed. In vision–language retrieval, ARO (Yuksekgonul et al., 2022) swaps two objects or two attributes in a caption, for example "the horse is eating the grass" to "the grass is eating the horse". It finds that models with strong retrieval scores often perform close to chance on these swaps. SugarCrepe (Hsieh et al., 2023) later showed that many such swapped captions can be rejected from the text alone, because they are implausible or ungrammatical. A test based on swapped descriptions must therefore check that the swap cannot be detected without looking at the input. D1 follows this approach for time series.

For time series, Tan et al. (2024) shuffle the inputs of forecasting models in three ways: the whole series, only its first half, or by swapping its two halves. They also mask part of the input. They find that forecasters built on language models are no more sensitive to order than the same forecasters with the language model removed or replaced by a simple layer. D2 adapts these perturbations to alignment. Ablations in time-series language models show that statistics written in the prompt can account for much of the performance (Yin et al., 2026), and that giving the series as text rather than through a time-series encoder changes which sub-tasks the model solves (Xie et al., 2024).

Finally, surrogate-data methods from nonlinear time-series analysis test which properties of a series an analysis depends on: they generate surrogate series that keep chosen properties, such as the distribution of values, and randomise the rest (Theiler et al., 1992). D3 applies the same idea to alignment models.

2.5 Positioning of this study

Table 2.1 compares this thesis with the closest prior work. Each prior study tests one or two of three kinds of change: to the description, to the order of the input, or to the numerical information the model receives. These studies also use other kinds of models or the authors' own models. This thesis applies all three kinds of change to the same existing time-series–text alignment models, and pairs each with checks that show whether the results can be interpreted, such as whether the model performs well enough before any change.

Table 2.1: **Comparison with prior work.** Which kinds of input change each study applies, and whether it includes checks on the evaluation itself. *Desc.*: changing one part of a description. *Order*: rearranging the values of the series. *Info.*: removing or replacing numerical information. *Evaluation checks*: tests of whether the results can be trusted, such as checking the metric, using a control model, or verifying the test items.

			Change to the input			Evaluation
Study	Models	Task	Desc.	Order	Info.	checks
ARO (Yuksekgonul et al., 2022)	vision–language	retrieval	✓	(✓)[a]		
SugarCrepe (Hsieh et al., 2023)	vision–language	retrieval	✓			✓[b]
Tan et al. (2024)	LLM-based forecasters	forecasting		✓		✓[c]
MMTS-Bench (Yin et al., 2026)	authors' time-series LLM	question answering			✓	✓[d]
ChatTS (Xie et al., 2024)	authors' time-series LLM	question answering			(✓)[e]	
This thesis	**time-series–text alignment**	**retrieval, two-choice**	✓	✓	✓	✓

[a] Rearranges the words of the caption, and in a separate retrieval analysis, the patches of the image. [b] Checks whether swapped captions can be rejected by reading the caption alone, without looking at the image. [c] Compares each forecaster with a version that has the language model removed or replaced by a simple layer. [d] The benchmark items are checked by people. [e] Gives the full series either as text or through the time-series encoder; the information itself is not reduced.

Chapter 3

Methodology

This chapter first defines the three diagnostics and the controls that make them interpretable, and then describes how they were applied: the models, datasets, item construction and statistical procedures.

3.1 Setting and notation

Each test item pairs a time series $\boldsymbol{x}$ with a natural-language description t. A series can have several channels (variables observed over the same time steps), and L denotes its number of valid time steps. The models are evaluated in one of two ways.

Retrieval. A retrieval model gives each series–description pair a similarity score. For each description used as a query, the model ranks all series in a candidate pool by this score. Let r_q be the rank of the correct series for query q. We report mean reciprocal rank (MRR), which gives more credit the higher the correct series is ranked,

$$\mathrm{MRR} = \frac{1}{Q}\sum_{q=1}^{Q}\frac{1}{r_q}, \tag{3.1}$$

where Q is the number of queries.

Two-choice. The model sees a series $\boldsymbol{x}$ and two descriptions, the correct one t^+ and an alternative, and selects one. A retrieval model selects the description with the higher score. A generative model ChatTS, which answers in text, is asked each question twice, with the two descriptions in swapped positions, and the two answers are averaged, so that a preference for one position cannot raise the score. Chance accuracy is 0.5.

Comparisons. Every diagnostic compares a model with itself on the same items, before and after a change to the input. For a performance measure P (MRR or accuracy), the change under a condition c and the relative loss are

$$\Delta_c = P_c - P_{\text{orig}}, \qquad \rho_c = \frac{P_{\text{orig}} - P_c}{P_{\text{orig}}}. \tag{3.2}$$

A loss therefore appears as a negative Δ_c and a positive ρ_c. For example, $\rho_c = 0.25$ means that a quarter of the original score was lost.

3.2 Diagnostic design

3.2.1 D1: Component sensitivity

A description is decomposed into *components*, such as trend direction or fluctuation type, each of which can take one of several options (for trend direction, for example, "rising" or "falling"). For a signal $\boldsymbol{x}$ with correct description t^+, a *swap item* replaces the wording of exactly one component of t^+ with a different option and leaves all other text unchanged, producing t^{swap}; a matched *random-control item* pairs t^+ with an unrelated description t^{rand} of another signal. We measure

$$\text{gap}_k = \text{Acc}\big(t^+ \text{ vs. } t^{\text{rand}}\big) - \text{Acc}\big(t^+ \text{ vs. } t^{\text{swap}_k}\big) \tag{3.3}$$

for each component k, where $\text{Acc}(t^+ \text{ vs. } t')$ is the share of items in which the model chooses t^+ over the alternative t'. The random control shows whether the model can match the signal at all. The gap measures how much of that ability is lost when the alternative differs from the correct description in component k only.

Because a swapped description shares most of its text with the correct one, it is usually harder to reject, so some positive gap is expected for most components. The informative pattern is therefore a component whose gap is clearly larger than the others *within* the same model. A swap is valid only if t^{swap} is genuinely false for $\boldsymbol{x}$; we check this by structural checks and human review (Section 3.7), and we test separately whether a swap can be rejected from its wording alone, without looking at the series (Section 4.2.1).

3.2.2 D2: Temporal-order sensitivity

D2 changes the series and leaves the description unchanged. Following Tan et al. (2024), we use three order perturbations and a masking condition (Figure C.1):

- *Full shuffle*: rearranges all time steps at random. It keeps every value but removes their order.
- *First-half shuffle*: rearranges only the first half.
- *Half exchange*: swaps the two halves and keeps the order within each half.
- *Masking*: replaces about one fifth of the time steps, chosen at random, with the series mean, which is zero after the model's own normalisation. It removes observations instead of moving them.

For multichannel series, all channels are rearranged in the same way, so the values observed at one time step stay together. Construction details are in Chapter C.

A drop in performance after shuffling does not show that the model uses order. A shuffled series may look unlike anything the model saw in training, so performance could fall even when the description is still true. We therefore split descriptions into two groups. A description is *order-dependent* if at least one of its claims can be made false by shuffling (e.g. "rises and then falls") and *order-invariant* if all of its claims remain true after shuffling (e.g. "the majority is flat"); descriptions that cannot be classified are excluded. If a model uses order, the order-dependent group should lose more than the order-invariant group. Writing Δ_{dep} and Δ_{inv} for the change in each group under full shuffling, we define the difference-in-differences

$$\text{DiD} = \Delta_{\text{inv}} - \Delta_{\text{dep}}, \tag{3.4}$$

which is positive when the order-dependent group loses more. Because the same DiD can arise from very different group changes, it is always reported together with both group changes and read according to Table 3.1.

Table 3.1: **Reading the difference-in-differences.** Readings assume sufficient original performance. "Little or no change" requires an equivalence test (Section 3.3); a non-significant difference is not enough.

Order-dependent	Order-invariant	DiD	Interpretation
large drop	little or no change	$+$	Performance falls where the description requires order: order-sensitive matching.
large drop	similar large drop	≈ 0	Shuffling harms both groups; the drop is not specific to order claims.
larger drop	smaller drop	$+$	Order claims are more affected, but matching also suffers where order is irrelevant.
little or no change	little or no change	≈ 0	Matching survives without order; D3 asks what supports it.
little or no change	improvement	$+$	The positive DiD comes from the invariant group improving, not from an order-specific loss.

3.2.3 D3: Statistical-information sufficiency

If some performance remains after shuffling, something other than order supports it. D3 identifies what by replacing the series with a *ladder* of inputs of the same length, each keeping less information than the one before (Table 3.2). The *shuffled* series keeps the exact values but not their order. The *own-value resample* draws new values at random, with replacement, from the original values, so it keeps their approximate distribution but not the exact values. The *matched Gaussian* draws values from a normal distribution with the series' mean and standard deviation, so it keeps only these two statistics and the length. For multichannel series, the resample draws whole time steps, so values observed at the same time stay together, while the matched Gaussian is drawn for each channel separately (Chapter C).

The diagnostic rests on three comparisons. (i) If performance holds from shuffled to resampled, the exact values are not needed. (ii) If it then falls from resampled to matched Gaussian, the *shape* of the value distribution, beyond its mean and standard deviation, was supporting it. (iii) For the retrieval models, if the matched Gaussian remains above chance, we test whether length alone explains it. We compare it with a *length-only reference*: the MRR of a ranker that knows only each candidate's length and ranks candidates of the same length at random (Chapter C). This reference equals chance when all candidates have the same length.

ChatTS additionally receives a five-field summary in place of the series: its mean, standard deviation, minimum, maximum and length, given as text. This is not part of the ladder; it tests whether ChatTS can answer when it is told these statistics directly.

Table 3.2: **The D3 ladder**: what each condition keeps from the original series.

Condition	Retained information
Original	everything (reference)
Shuffled	the exact multiset of values; not their order
Own-value resample	the empirical value distribution; not the exact values
Matched Gaussian	mean, standard deviation and length only
Five-field summary (ChatTS only)	mean, s.d., min, max and length, as text; no series

What a condition actually leaves to a model depends on the model's preprocessing. We state this per model in Section 3.6. If performance remains meaningful under a reduced input, that information can support matching, but this does not show that the model used it to produce its original score. If performance falls to chance under every reduction, only the tested reductions are ruled out.

3.3 Validity controls and verdicts

Each diagnostic compares two scores, so a misleading score leads to a misleading result. We use five controls to prevent this.

V1 Metric audit. Before interpreting changes in a metric, we check that it separates a trained model from the same architecture with random weights. A metric that fails this cannot detect the failures the diagnostics are designed to provoke.

V2 Capability gating. A model with little reference performance has little to lose. Each model–dataset pair (and, in D1, each component's random control) receives a capability label: *viable* when it clearly supports interpretation, *weakly viable* when above chance but limited, and VOID when too weak to interpret. For accuracy, the guide is whether the lower end of the 95% interval exceeds 0.60; for retrieval, whether MRR lies clearly above chance for the pool. VOID results are reported but not interpreted.

V3 Item certification. A swapped description must really be false for the signal, and each description must be placed in the right group (order-dependent or order-invariant). These items are checked by hand. Section 3.7 states which sets were checked in full and which by sample.

V4 Equivalence testing. A claim of "little or no change" requires two one-sided tests (TOST) (Schuirmann, 1987; Lakens, 2017): the 90% bootstrap interval of the difference must lie within ± 0.05. Outcomes are *equivalent* (interval inside the margin), *shifted* (interval excludes zero and extends beyond the margin) or *inconclusive* (interval includes zero and extends beyond the margin). Because the margin is absolute, a model whose original score is below 0.05 passes even if it loses everything; we flag every such case.

V5 Negative control. We also evaluate a general text-embedding model, text-embedding-3-large, which has no time-series encoder. Each series is written as a list of numbers in text, and the model scores how similar this text is to each description. In D1, the description with the higher similarity is counted as its choice; in D2 and D3, all series are ranked by similarity, as for the retrieval models. Because this model has almost no matching ability, a pattern it produces cannot reflect matching between series and description. A main-model pattern that it reproduces is therefore not, by itself, evidence about the main model; such a pattern can arise, for example, from a small test group or from how the test items are built.

3.3.1 Verdicts

The results report a short label, or *verdict*, for each D1 component and each D3 setting. This section defines when each label is given.

D1. A component receives the label COLLAPSE or PARTIAL only if its swap accuracy is significantly below its random control, consistently across runs where runs exist. It is COLLAPSE when the lower end of the swap accuracy's 95% interval falls below 0.60, and PARTIAL otherwise. COLLAPSE does not mean chance-level performance.

D2. D2 has no single label. It is summarised by the size of each perturbation effect, the DiD together with its two group changes, and the equivalence outcomes.

D3. We assign SHORTCUT PRESENT when information that survives without the original order supports matching above the relevant reference level, and NO SHORTCUT DETECTED when this is not shown under the tested reductions. The label is operational: it marks information that supports matching without order, whether or not that information is legitimate for a given description. When a model is already at chance after shuffling there is no residual, and the ladder can only confirm that no reduced input restores performance.

3.4 Models

We evaluate three time-series–text alignment models and one negative control (Table 3.3). Apart from training our CLaSP reimplementation, no model is trained or fine-tuned for this study.

CLaSP (Ito et al., 2025) is a retrieval model with two encoders, one for the series and one for the text, trained so that matching series–description pairs receive high similarity. It has no official code, so we reimplemented it and trained it jointly on SUSHI and TRUCE with three seeds (42, 43, 44). The one deliberate deviation from the paper is the series encoder: we replace the Informer with a standard Transformer encoder, which is still tractable at the maximum length of 2,048 points (Chapter A). Our findings therefore describe this reimplementation.

TRACE (Chen et al., 2026) is a retrieval model of the same kind, designed for series with several variables and longer descriptions. We use the checkpoint released by the authors, trained on weather data, with its original Nomic text encoder (Nussbaum et al., 2024).

On the released test data, its MRR is close to the published values. TRACE hides a random 30% of its input at every evaluation; we keep this mask the same across conditions and treat three different masks as three runs.

ChatTS-14B (Xie et al., 2024) is a large language model that reads a time series directly alongside a text prompt. We give it a series and two candidate descriptions and ask it to choose one, in both answer orders. We read the answer from the model's scores for the two options, which matched its written answers in all checked cases. Before encoding, ChatTS subtracts the series mean, rescales the series only when its values lie far from the mean, cuts it into 16-point patches, and writes the removed mean and scale into the prompt as numbers. Our conclusions apply to this version and question format, not to ChatTS's free-form answers.

TE-3L (negative control). `text-embedding-3-large` is a general-purpose model that converts text into vectors. It has no time-series encoder. Each series is standardised, multiplied by 10, rounded and written as comma-separated integers. The model embeds this text and the descriptions separately, and their similarity is used as the matching score. All time points are kept, at reduced precision. We refer to its score as the *embedding floor*: the score reachable without a time-series encoder.

Finally, an **untrained CLaSP** (same architecture, random weights) is used only for the metric audit (V1).

Table 3.3: **Systems.** "Runs" states what varies across repeated runs.

System	Series input and preprocessing	Runs
CLaSP (reimpl.)	Transformer encoder; per-series standardisation	3 training seeds
TRACE (released)	7 channels, padded to 186; per-channel standardisation; built-in 30% mask	3 mask draws
ChatTS-14B	patch encoder; mean and scale removed and stated in the prompt	1 (both orders)
TE-3L (control)	series written as integer text	1 (D1); 3 (D2, D3)

3.5 Datasets

We use three datasets that pair time series with text (Table 3.4); examples are in Section B.1.

SUSHI (Kawaguchi et al., 2024) is a synthetic dataset. We use its public Tiny release of 1,400 series of 2,048 points. Each of its 140 classes combines a global shape (e.g. linear decrease,

sinusoidal wave) with a fluctuation type (e.g. negative spikes, noise), and each series has one caption built from templates with some manual editing. A class-stratified 8:1:1 split gives 1,120 training, 140 validation and 140 test series.
TRUCE (Jhamtani and Berg-Kirkpatrick, 2021) contains synthetic and stock-price series of 12 points, each with three captions written by people. We keep the released splits, with 56 synthetic and 190 stock series in the test set.
NOAA is the weather dataset released with TRACE. Each record holds seven variables (temperature, precipitation, relative humidity, visibility, eastward and northward wind, and sky cover) over 102–180 time steps, padded to 186. Its generated description has ten text fields (time span, location, weather labels, and one field per variable), which are joined into a single description for retrieval.

Table 3.4: **Datasets**

Dataset	Series	Length	Descriptions	Test data
SUSHI Tiny	synthetic, univariate	2,048	templated, 1 per series	140 series / 140 captions
TRUCE	synthetic + stock, univariate	12	human, 3 per series	246 series / 738 captions
NOAA (TRACE)	weather, 7 channels	102–180	generated, 10 fields	2,006 pairs

3.6 Instantiating the diagnostics

Table 3.5 shows how each model is evaluated on each dataset. In D1, every model is evaluated in the two-choice format of Section 3.1: it chooses between the correct description and either a copy with one component swapped or a random description. In D2 and D3, CLaSP and TE-3L rank the correct series among the 386 merged SUSHI and TRUCE test series (chance MRR 0.017), and TRACE among 2,006 NOAA test series (chance 0.004). In the merged pool, length alone separates the two datasets (2,048 vs. 12 points); Section 4.4 shows that this matters. In D2 and D3, ChatTS instead answers a two-choice question. For each caption, the wrong option is an unrelated description and stays the same across all conditions. Table 3.6 lists the conditions run for each model–dataset pair, and results in Chapter 4 are reported per pair.

Table 3.5: **Experimental setup.** How each model is evaluated on each dataset.

Model	Dataset	Evaluation	Candidate setting	Diagnostics
CLaSP	SUSHI	two-choice (accuracy)	2 descriptions	D1
	SUSHI, TRUCE	retrieval (MRR)	386 merged series	D2, D3
TRACE	NOAA	two-choice (accuracy)	2 descriptions	D1
	NOAA	retrieval (MRR)	2,006 series	D2, D3
ChatTS	SUSHI	two-choice (accuracy)	2 descriptions	D1
	SUSHI, TRUCE	two-choice (accuracy)	2 descriptions	D2, D3
TE-3L (control)	SUSHI	two-choice (accuracy)	2 descriptions	D1
	SUSHI, TRUCE	retrieval (MRR)	386 merged series	D2, D3

D1 on SUSHI. We derive five components from the class labels (Table 3.7): C1, C2, C3 and C5 come from the shape part of the label, C4 from the fluctuation part. A swap edits the caption, not the label: for C1–C3 and C5 the shape sentence is replaced by a real caption sentence from the target category (chosen to be close in length) and the fluctuation sentence is kept; for C4, the fluctuation sentence is replaced and the shape sentence is kept.

Table 3.6: **Experimental coverage.** "Same four/three" refers to the conditions in the first row.

Model on data	D1 components	D2 perturbations	D3 conditions
CLaSP on SUSHI	C1–C5 (C4: 738 cert.)	full shuffle, first-half shuffle, half exchange, masking	full shuffle, own-value resample, matched Gaussian
CLaSP on TRUCE	—	same four	same three
TRACE on NOAA	N1, N2, N3 (344 cert.), N5	same four	same three
ChatTS on SUSHI	C1–C5 (C4: 738 cert.)	same four; within-patch and patch-order shuffle; prompt-number control	same three; five-field summary; two series/prompt-number controls
ChatTS on TRUCE	—	same four; prompt-number control	same three; five-field summary; two controls
TE-3L on SUSHI	C1–C5 (C4: all 990)	same four	same three
TE-3L on TRUCE	—	same four	same three

Sampling from the 280 validation and test series gives 5,540 questions, half swaps and half random controls, which together use 279 of the series. For C4, filtering and full human review leave 738 certified swaps (Section 3.7).

D1 on NOAA. SUSHI series could not be used for TRACE: shortened to TRACE's maximum input length, they lose the difference between spike types. A classifier on hand-crafted signal features, which separates the three spike types at full length, falls to near chance after shortening (Section B.2). A failure of TRACE on such data could not be attributed to TRACE, so TRACE is tested on its own data. We derive four components from four of the ten fields (Table 3.7); each swap changes one field, leaves the other nine fields and all seven series unchanged, and is joined into a full description. Location (N5) does not appear in the series and was intended as a component that should *not* be detectable. N3 is evaluated on 344 certified swaps. A planned component N4 (fluctuation vs. stability) was dropped because only 33 records admitted a consistent edit.

D2 grouping. A rule-based classifier assigns each description to the order-dependent, order-invariant or unclear group; every order-invariant and unclear description, and samples of order-dependent ones, were checked by hand (Section B.4). Order-invariant descriptions are rare (Table 3.8): SUSHI has 4, the TRUCE test set 18 (over 14 series), and NOAA none, so a DiD is reported on TRUCE, descriptive only on SUSHI, and impossible for TRACE. SUSHI's single flat test series, which shuffling cannot change, is left out of both groups and used only as a check. For TRACE the halves and permutations are formed over valid steps only; padding stays in place. For ChatTS on SUSHI we add two patch-level shuffles (within each 16-point patch, and of the patch order), and on both datasets a *prompt-number control* that shifts only the two prompt numbers by the amount masking would, leaving the series untouched.

D3: what each condition leaves to each model. CLaSP, TRACE and TE-3L standardise every series (or channel) to mean 0 and standard deviation 1. For these models, the matched Gaussian therefore keeps little beyond the length of the series. For TRACE, resampling draws whole time steps and copies all seven values, whereas the Gaussian draws each channel

Table 3.7: **D1 components.** Top: derived from SUSHI class labels. Bottom: derived from NOAA text fields. † Evaluated on human-certified swaps.

Component	What the swap changes	Example
C1 trend direction	reverses the shape's direction, keeps the curve type	linear decrease → linear increase
C2 trend family	changes the curve type (linear, exponential, sigmoid, Gaussian, quadratic, cubic), keeps direction	linear decrease → exponential decay
C3 periodic waveform	changes the wave shape of a periodic signal	sinusoidal → square wave
C4 fluctuation type†	changes the fluctuation (smooth, noisy, ±spikes, step), keeps the shape	negative spike → positive spike
C5 signal regime	replaces a trend by a periodic shape or vice versa	linear increase → sawtooth wave
N1 weather labels	replaces every label that has an opposite (Hot/Cold, Warm/Cool, Clear/Cloudy, Rainy/Dry)	Clear, Warm, Dry → Cloudy, Cool, Rainy
N2 time span	changes the stated span (one week ↔ six months) and adjusts dates	the past week → the past six months
N3 temperature trend†	swaps warming and cooling words; stated values unchanged	upward trend → downward trend
N5 location	replaces the location field by one from another US state	Polk, Iowa → Ocean, New Jersey

Table 3.8: **Description groups** in the test sets used by D2.

Dataset	Order-dependent	Order-invariant	Excluded	DiD
SUSHI	135	4	1 constant signal	descriptive only
TRUCE	715 (245 series)	18 (14 series)	5 unclear	reported
NOAA	2,005	0	1 unclear	not computable

independently; the resample-to-Gaussian step therefore removes *both* the per-channel distribution shape *and* cross-channel co-occurrence, and the design cannot separate them. TRACE's padding and its constant channels survive every condition. ChatTS does not standardise, so its Gaussian still carries level and spread through the prompt numbers; two further controls pair a Gaussian series with the original prompt numbers, and the original series with another series' numbers. ChatTS additionally receives the five-field summary in place of the series. Full construction details are in Chapter C.

3.7 Item certification

All human checks were made by a single annotator (the author). A swap is counted as valid only if the edited description is well formed, states the new option, and is clearly false for the signal to a competent reader. Two components were checked item by item: of 990 generated SUSHI fluctuation-type swaps (C4), 738 remain after automatic filtering and review, and of 400 generated NOAA temperature-trend swaps (N3), 344 remain. The other components passed automated structural checks; for SUSHI, a small sample from each component was also checked by hand (Chapter B). NOAA's N1, N2 and N5 were not checked by hand, and neither were random-control descriptions. Removal reasons and examples are given in

Chapter B.

3.8 Statistical procedures and recorded predictions

Uncertainty is shown with 95% bootstrap intervals (Efron and Tibshirani, 1994) that resample signals, since one signal may have several descriptions. Conditions are compared with paired tests on the same items: a paired bootstrap and the Wilcoxon signed-rank test (Wilcoxon, 1945), with McNemar's test (McNemar, 1947) as a second check for ChatTS in D1. When several components are tested at once, the main p-values are adjusted with the Holm method (Holm, 1979). Effects must also replicate across runs where runs exist. The equivalence margin of ± 0.05 (V4) is based on the between-seed variability of CLaSP (Section 4.1), and its meaning per comparison is given in Table 3.9. Two choices were made after seeing some results: the margin was fixed after the D1 point estimates had been seen, and the length-only reference was adopted after the CLaSP D3 results. Predictions recorded before each analysis, and their outcomes, are reported in Chapter G.

Table 3.9: **Meaning of the** ± 0.05 **equivalence margin** for each comparison.

Comparison	Meaning of ± 0.05
D1 accuracy gaps; ChatTS accuracy comparisons	5 percentage points of accuracy
CLaSP D2 invariant-group change; TE-3L D2 and D3 changes	0.05 MRR
CLaSP and TRACE D3: resample vs. shuffle	0.05 difference in relative MRR loss

Chapter 4

Results

This chapter first checks whether the results can be interpreted (Section 4.1) and then answers the three research questions in turn (Sections 4.2 to 4.4). Each of these three sections has the same structure: a short summary of the design, the main findings, checks and interpretation, and the answer to the research question.

4.1 Reference capability and metric validity

Two conditions have to hold before any diagnostic result can be interpreted. A model must perform well enough on the unaltered input to have something to lose, and the evaluation metric must be able to show a failure when one occurs.

Reference capability (V2). All three alignment models perform above chance on the unaltered inputs (Table 4.1). CLaSP and TRACE are far above chance on every dataset. ChatTS is viable on SUSHI and weakly viable on TRUCE, where the lower end of its interval (0.590) lies just below the 0.60 threshold; its TRUCE results are therefore read with more caution. The TE-3L floor is VOID on both datasets, as expected for a negative control. Values can only be compared within a row.

Table 4.1: **Reference performance for D2 and D3.** Retrieval models rank the correct series among 386 merged SUSHI +TRUCE series or 2,006 NOAA series. Ranges span three CLaSP seeds or three TRACE mask draws; brackets are 95% intervals. D1 capability is assessed per component (Table 4.3).

Model	Data	Measure	Queries	Value	Chance	Label
CLaSP	SUSHI	MRR	140	0.304–0.368	0.017	viable
CLaSP	TRUCE	MRR	738	0.099–0.107	0.017	viable
TRACE	NOAA	MRR	2,005	0.552–0.564	0.004	viable
ChatTS	SUSHI	accuracy	140	0.743 [0.682, 0.807]	0.5	viable
ChatTS	TRUCE	accuracy	738	0.622 [0.590, 0.653]	0.5	weakly viable
TE-3L	SUSHI	MRR	140	0.004	0.017	VOID
TE-3L	TRUCE	MRR	738	0.032	0.017	VOID

Metric audit (V1). The judge-based mAP@10 used to report CLaSP fails this check in one of its four configurations. In this metric, a retrieved series counts as relevant whenever Sentence-BERT or DistilBERT (Sanh et al., 2019) rates its caption as similar enough to the query (Ito et al., 2025). We computed it for the trained and the untrained CLaSP under the

four judge configurations of the paper (Table 4.2). With DistilBERT at a threshold of 0.5, the untrained model scores 0.999 on TRUCE and the trained model 0.997, and both score 1.000 on SUSHI. The judge accepts almost any retrieved series, so in this configuration the metric does not reflect retrieval quality. The other three configurations do separate the two models, so the problem is specific to this judge and threshold. To avoid relying on a judge at all, all further retrieval results use strict retrieval: only the series paired with the query counts as correct, and we report MRR (Section 3.1). This metric passes the same check: the trained CLaSP reaches an MRR of 0.132–0.149 over all test queries, the untrained model 0.010 (chance 0.017).

Table 4.2: **Metric audit.** Judge-based mAP@10 for trained CLaSP (seed 42) and the same architecture with random weights; "paper" gives the values reported by Ito et al. (2025). Our values rank against the merged 386-series pool and the public SUSHI Tiny release, so they are not directly comparable with the paper's (Section A.1).

		TRUCE			SUSHI		
Judge	Threshold	trained	untrained	paper	trained	untrained	paper
Sentence-BERT	0.5	0.440	0.277	0.458	0.854	0.083	0.982
Sentence-BERT	0.8	0.105	0.017	0.136	0.306	0.001	0.571
DistilBERT	**0.5**	**0.997**	**0.999**	1.000	**1.000**	**1.000**	1.000
DistilBERT	0.8	0.550	0.378	0.491	0.932	0.051	0.992

4.2 D1: Which description components are models sensitive to?

Figure 4.1 and Table 4.3 compare swap accuracy with random-control accuracy for every component. Every system with interpretable results shows a significant drop on at least one component, and the weakest component differs by model.

CLaSP on SUSHI: fluctuation type. Trend direction, trend family and periodic waveform show a PARTIAL drop at all three checkpoints. Fluctuation type shows the largest gap. Swap accuracy falls to 0.592–0.618 against 0.970–0.973 in the random control, a gap of 0.355–0.378 that is significant at every checkpoint (Holm-adjusted $p < 0.001$), with swap lower bounds of 0.545–0.570. Because these lower bounds fall below 0.60, fluctuation type is labelled COLLAPSE, although CLaSP still does better than guessing on these swaps. Signal regime is the only component on which CLaSP does better on the swap than in the random control (0.980–0.989 vs. 0.949–0.957). One possible explanation is that the swap replaces a trend with a periodic shape, or the reverse, so the wrong description is very different from the correct one. A random description, by contrast, can describe a similar shape: a linear increase and exponential growth, for example, both describe rising patterns.

TRACE on NOAA: temperature-trend direction. All four components show a significant PARTIAL drop under all three mask draws. Temperature-trend direction has the largest gap (0.282–0.297): swap accuracy falls to 0.703–0.712, compared with 0.994–1.000 for the random-control. Despite this large drop, it is labelled PARTIAL rather than COLLAPSE because its swap lower bounds (0.654–0.666) remain above 0.60. Location is a different case. None of the seven weather channels in series explicitly records location, so a location swap was not expected to be detectable from the series, yet TRACE's accuracy on location swaps is 0.918–0.935 (Section 4.2.1).

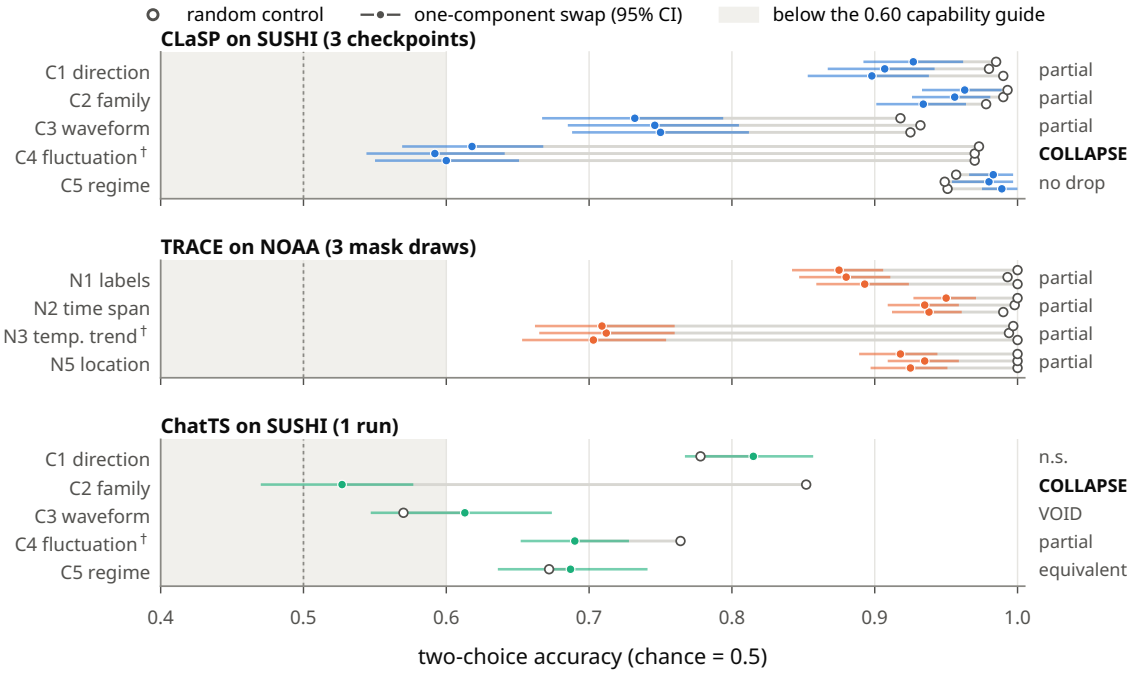


Figure 4.1: **D1: random-control vs. one-component swap accuracy.** Each line connects a component's random-control accuracy (open) to its swap accuracy (filled, 95% interval); for CLaSP and TRACE the three runs are drawn separately. The shaded band marks swap accuracies whose interval can fall below the 0.60 capability guide that separates COLLAPSE from PARTIAL. † ,Only swaps that passed human review are used..

Table 4.3: **D1 results.** Gap = random-control − swap accuracy. Top: ranges over three runs; "Swap LB" is the lower end of the swap accuracy's 95% interval in each run. Bottom: one run with 95% signal-bootstrap intervals. † Human-certified items. Per-run values and tests: Chapter D. Compare within a model only.

Model	Component	Random	Swap	Gap	Swap LB	Items/signals	Verdict
CLaSP on SUSHI	C1 trend direction	.980–.990	.898–.927	+.059 to +.093	.854–.893	205/205	PARTIAL
	C2 trend family	.978–.993	.934–.963	+.029 to +.044	.902–.934	410/177	PARTIAL
	C3 periodic waveform	.918–.932	.732–.750	+.175 to +.186	.668–.689	280/70	PARTIAL
	C4 fluctuation type†	.970–.973	.592–.618	+.355 to +.378	.545–.570	738/237	**COLLAPSE**
	C5 signal regime	.949–.957	.980–.989	−.037 to −.026	.955–.976	885/248	no drop (swap higher)
TRACE on NOAA	N1 weather labels	.993–1.00	.875–.893	+.108 to +.125	.843–.860	400/400	PARTIAL
	N2 time span	.990–1.00	.935–.950	+.050 to +.063	.910–.928	400/400	PARTIAL
	N3 temperature trend†	.994–1.00	.703–.712	+.282 to +.297	.654–.666	344/344	PARTIAL
	N5 location	1.00	.918–.935	+.065 to +.083	.890–.910	400/400	PARTIAL

Model	Component	Random	Swap	Gap	Items/signals	Verdict
ChatTS on SUSHI	C1 trend direction	.778 [.724, .827]	.815 [.768, .856]	−.037 [−.100, +.022]	205/205	no sig. drop
	C2 trend family	.852 [.821, .883]	.527 [.471, .576]	+.326 [+.275, +.380]	410/177	**COLLAPSE**
	C3 periodic waveform	.570 [.502, .638]	.613 [.548, .673]	−.043 [−.125, +.038]	280/70	VOID
	C4 fluctuation type†	.764 [.729, .796]	.690 [.653, .727]	+.075 [+.034, +.116]	738/237	PARTIAL
	C5 signal regime	.672 [.624, .718]	.687 [.637, .740]	−.015 [−.056, +.024]	885/248	equivalent
TE-3L on SUSHI	C1 trend direction	.663 [.595, .727]	.576 [.512, .644]	+.088 [+.005, +.171]	205/205	VOID
	C2 trend family	.654 [.597, .708]	.446 [.390, .505]	+.207 [+.148, +.267]	410/177	VOID
	C3 periodic waveform	.221 [.157, .289]	.532 [.461, .600]	−.311 [−.386, −.236]	280/70	VOID
	C4 fluctuation type	.553 [.507, .600]	.472 [.426, .518]	+.081 [+.025, +.137]	990/240	VOID
	C5 signal regime	.405 [.353, .459]	.513 [.446, .585]	−.108 [−.158, −.062]	885/248	VOID

ChatTS on SUSHI: trend family. Trend family shows the largest drop, from 0.852 in the random control to 0.527 on the swaps. The whole swap interval $[0.471, 0.576]$ lies below 0.60 and includes chance, so the component is labelled COLLAPSE: ChatTS cannot reliably tell trend families apart. Fluctuation type shows a PARTIAL drop ($0.764 \rightarrow 0.690$). Trend direction shows no significant drop, but the interval is too wide to show that the two conditions are the same. Signal-regime swaps are as hard as the random control (the two are equivalent). Periodic waveform is VOID: even against random descriptions ChatTS reaches only 0.570 (lower bound 0.502), so its swap result cannot be interpreted.

Negative control and capability gating. All five TE-3L components are VOID: three random controls have lower bounds between chance and the 0.60 capability guide, and two lie clearly *below* chance (0.221 and 0.405), so random alternatives are not neutral for a text-only model. Yet all five gaps are statistically significant (three positive, two negative). Without the reference-capability check (Section 4.1), these gaps would be reported as component effects of a model that cannot match series at all.

4.2.1 Robustness and limitations of D1

The collapses are not caused by missing information in the signal. A collapse would say little about a model if the two options of a swap could not be told apart from the signal itself. We tested this directly: for both collapsed components, the two options can be told apart from the signal alone. A logistic regression on 16 hand-crafted shape and texture features of the standardised signal (Section D.1) separates the two values of each swapped pair with 5-fold cross-validated accuracy of 0.92–0.99 per component. Fluctuation type (0.93) and trend family (0.94), the two collapsed components, fall within the same range as the others. The only pair the features cannot separate is sinusoidal versus triangle wave (0.46), which belongs to periodic waveform, not to a collapsed component. The collapses therefore cannot be explained by missing information in the signal.

The models do not simply pick descriptions by length. In principle, a model could choose between two descriptions by their length alone, without using the series at all. This would work if correct and wrong descriptions systematically differed in length. Two checks show that the models do not do this. First, choosing by length alone would be barely better than guessing on SUSHI and far worse than TRACE on NOAA. A rule that always picks the longer (or always the shorter) description scores at most 0.52 on SUSHI swaps and 0.58 on SUSHI random controls, and at most 0.60 on NOAA swaps (time span) and 0.75 on NOAA random controls (weather labels), compared with 0.935–0.950 and 0.993–1.000 for TRACE on those items. Second, the choices of CLaSP and TRACE do not follow length at all: across swap items, how much longer one description is has almost no relation to how strongly the model prefers it (correlation 0.02 for CLaSP and ≈ 0.00 for TRACE). For weather labels and temperature-trend direction, length offers no cue anyway, because the two descriptions always have the same number of words. The D1 results therefore do not come from description length.

Temperature-trend swaps are harder for TRACE on one-week records. Among the certified temperature-trend swaps, accuracy is 0.600–0.644 on 90 records spanning one week, compared with 0.731–0.747 on 245 records spanning 28 days. The one-week group scores lower under all three mask draws. Random-control accuracy remains at 0.99–1.00 for both groups, showing that TRACE can still distinguish correct descriptions from random alternatives at either time span. TRACE's ability to detect a reversed temperature trend therefore differs between records of different time spans. However, the two groups are different sets

of records, so the difference could also come from other information in the records, not only from the time span.

The source of TRACE's location accuracy remains unclear. The location name cannot influence the series embedding used for retrieval, which is computed only from the seven weather channels (Section D.1). Changes to other wording are not required either: on the 40 swaps that change only the place name, accuracy is still 0.900 under all three mask draws. Three possible explanations remain. First, the standardised weather series may retain location-related patterns, such as the shape of daily variation. Second, TRACE may have learned associations with locations seen during training. Third, swapping a location may make the ten-field text *internally inconsistent*, allowing the model to favour the original description without relying on the matching series. Similar text-based shortcuts have been reported for vision–language hard negatives (Hsieh et al., 2023). This concern also applies to temperature-trend swaps, which reverse the described trend but keep the stated temperatures. Testing whether the model still favours the original description when both are scored against unrelated series would help assess this possibility. Location therefore cannot serve as the undetectable component it was intended to be.

4.2.2 Answer to RQ1

Every model with interpretable results loses much of its advantage over random alternatives on at least one component: fluctuation type for CLaSP (gap ≈ 0.36, COLLAPSE), trend family for ChatTS (0.33, COLLAPSE, swap accuracy not above chance) and temperature-trend direction for TRACE (≈ 0.29, PARTIAL). For both collapsed components, a simple classifier using basic features of the signal can tell the two options apart from the signal alone (accuracy ≥ 0.93; Section 4.2.1), so the collapses are not caused by missing information: the models fail to pick up a difference that the signal contains. These weaknesses are not visible in aggregate scores. Revealing them requires a component-level diagnostic such as D1.

4.3 D2: How much does matching depend on temporal order?

Figure 4.2 and Table 4.4 show that full shuffling removes most above-chance performance on order-dependent descriptions for every interpretable model. In Figure 4.2, higher bars in the retrieval panel (left) indicate a larger share of MRR lost, while bars extending further below zero in the ChatTS panel (right) indicate a larger drop in accuracy. Table 4.5 separates the two description groups wherever order-invariant descriptions exist.

CLaSP on SUSHI: large losses under shuffling. On order-dependent descriptions, full shuffling removes 76–78% of MRR, leaving 0.070–0.078, still above chance (0.017). Half exchange is almost as damaging (68–76%), despite preserving order within each half. First-half shuffling removes only 15–25%, while masking removes 15–48%, with large variation across checkpoints.

CLaSP on TRUCE: order-specific losses at two checkpoints. On order-dependent descriptions, full shuffling removes 66–75% of MRR but leaves performance above chance. Half exchange removes even more (80–84%), bringing MRR close to chance. Masking only two of the 12 points has no significant effect. At checkpoints 43 and 44, order-dependent descriptions lose MRR while order-invariant descriptions show no meaningful change ($\mathrm{DiD} \approx +0.075$, intervals above zero), consistent with an order-specific loss (Table 4.5). At checkpoint 42,

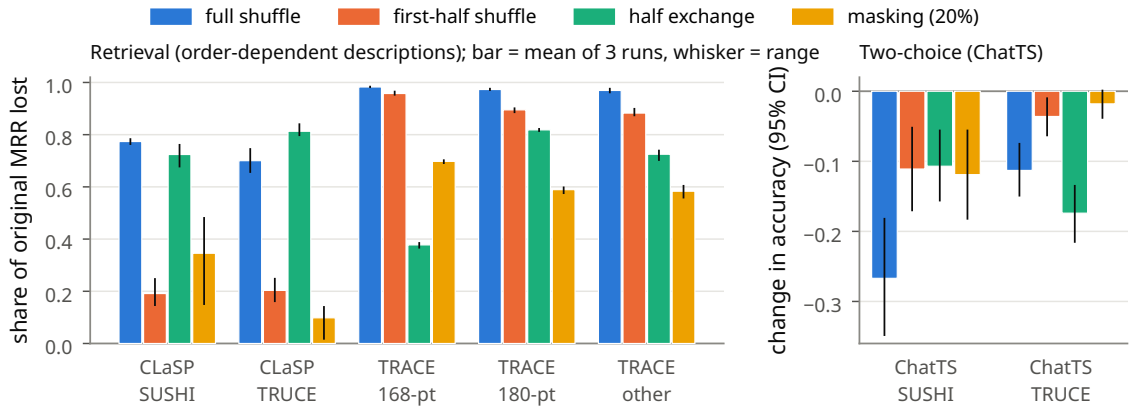


Figure 4.2: **D2: effect of each order perturbation on order-dependent descriptions.** Left: share of the original MRR lost by the retrieval models (bar: mean of three runs; whisker: range). Right: change in ChatTS accuracy with 95% intervals. Metrics differ between panels and are not comparable across them.

Table 4.4: **D2 on order-dependent descriptions.** Top: share of the original MRR lost (0: none, 1: all), and MRR remaining after full shuffling; chance MRR is 0.017 (CLaSP) and 0.004 (TRACE). Ranges over three runs. Bottom: change in ChatTS accuracy (one run, 95% intervals; chance 0.5). † The 95% interval of the MRR change includes zero in at least one run. ‡ Equivalent to no change. TRACE length groups are descriptive point estimates; pooled over all 2,005 descriptions, every TRACE change is significant (Chapter E).

Model on data	Desc.	Orig. MRR	Full shuffle	MRR after	First-half	Half exch.	Masking
CLaSP on SUSHI	135	.304–.358	.764–.783	.070–.078	.147–.247†	.678–.761	.151–.481
CLaSP on TRUCE	715	.100–.110	.657–.745	.028–.034	.162–.248	.798–.840	.018–.140†
TRACE, 168-pt	1,050	.579–.582	.982–.984	.009–.010	.953–.965	.367–.385	.691–.702
TRACE, 180-pt	544	.551–.572	.971–.976	.013–.016	.886–.901	.814–.822	.576–.598
TRACE, other	411	.478–.506	.962–.976	.012–.019	.874–.899	.703–.739	.559–.604

Model on data	Orig. accuracy	Full shuffle	First-half	Half exch.	Masking
ChatTS on SUSHI (135)	.733 [.670, .796]	−.267 [−.348, −.182]	−.111 [−.170, −.052]	−.107 [−.156, −.056]	−.119 [−.182, −.056]
ChatTS on TRUCE (715)	.624 [.593, .655]	−.113 [−.149, −.075]	−.036 [−.063, −.010]	−.174 [−.215, −.135]	−.018‡ [−.038, +.001]

Table 4.5: **D2 group comparison under full shuffling.** Δ: shuffled minus original (negative = loss); $\mathrm{DiD} = \Delta_{\mathrm{inv}} - \Delta_{\mathrm{dep}}$. 95% intervals in brackets. "Driven by" states which group change produces the DiD; at CLaSP checkpoints 43–44 the invariant group is equivalent to no change, and for ChatTS on TRUCE about two-thirds of the DiD comes from the invariant group's rise. SUSHI has four order-invariant descriptions, so its rows are descriptive. Per-run values: Chapter E.

		Order-dependent		Order-invariant			
Model	Run	Orig.	Δ_{dep}	Orig.	Δ_{inv}	DiD	Driven by
Retrieval (MRR)							
CLaSP, TRUCE	ckpt 42	.100	−.066 [−.085, −.047]	.077	−.064 [−.199, −.003]	+.002 [−.128, +.072]	both groups drop
	ckpt 43	.108	−.075 [−.096, −.057]	.021	+.001 [−.014, +.021]	+.076 [+.051, +.106]	dependent drop
	ckpt 44	.110	−.082 [−.100, −.064]	.024	−.009 [−.020, +.002]	+.073 [+.052, +.094]	dependent drop
CLaSP, SUSHI	3 ckpts	.304–.358	−.280 to −.232	.375–.667	−.138 to +.108	+.102 to +.340	descriptive only
TE-3L, TRUCE	draw 42	.030	−.010 [−.018, −.002]	.071	+.001 [−.046, +.046]	+.010 [−.042, +.057]	neither changes
	draw 43	.030	−.010 [−.018, −.003]	.071	+.073 [+.006, +.176]	+.084 [+.014, +.186]	**invariant rise**
	draw 44	.030	−.007 [−.015, +.000]	.071	+.065 [+.001, +.168]	+.072 [+.008, +.187]	**invariant rise**
Two-choice (accuracy)							
ChatTS, SUSHI	1 run	.733	−.267 [−.348, −.182]	1.00	.000 [.000, .000]	+.267 [+.186, +.351]	descriptive only
ChatTS, TRUCE	1 run	.624	−.113 [−.149, −.075]	.528	+.222 [+.053, +.429]	+.335 [+.143, +.550]	mostly invariant rise

both groups lose about the same amount (DiD ≈ 0). Most (87%) of the order-invariant group's loss comes from one description, "The majority is flat.", whose correct series falls from rank 1 to rank 259 even though the description remains true after shuffling.

TRACE on NOAA: large losses under shuffling. All 2,005 descriptions are order-dependent, so there is no order-invariant group for a DiD comparison. Full shuffling removes 96–98% of MRR across length groups, and first-half shuffling is almost as damaging (87–97%). After full shuffling, a small amount of performance remains above chance (MRR 0.012–0.013 vs. chance 0.004). Half exchange causes less loss than masking for 168-point series (37–39% vs. 69–70%), but more loss for 180-point series (81–82% vs. 58–60%). This occurs in all three draws, and the difference between the groups is significant in every draw. Pooling the groups would hide it. The 168-point series are one-week or 28-day records, and the 180-point series with a known span are six-month records. One possible explanation, not tested here, is that short records are dominated by repeating daily cycles, which survive a half exchange, whereas six-month records follow a seasonal progression, which the exchange disrupts (Chapter E).

ChatTS on SUSHI: patch order matters. Full shuffling lowers accuracy from 0.733 to 0.467, a level not distinguishable from chance. Each of the other perturbations reduces accuracy by about 0.11. Two additional tests examine order at the level of the 16-point patches that ChatTS reads. Shuffling points within each patch is equivalent to no change, while shuffling the order of the patches reduces accuracy by 0.219, close to the loss under full shuffling (Chapter E). These results suggest that ChatTS relies on the order of patches, but not on point order within them, when answering these questions.

ChatTS on TRUCE: accuracy falls to chance. ChatTS is only weakly viable on TRUCE (Section 4.1). Full shuffling lowers accuracy from 0.624 to 0.511, reaching chance, while half exchange reduces it further to 0.450, below chance. Masking is equivalent to no change. The large DiD (+0.335) comes mainly from the 18 order-invariant descriptions improving from 0.528 to 0.750. The order-dependent group itself loses 0.113, providing more direct evidence of sensitivity to order.

TE-3L: little performance to lose. On order-dependent descriptions, TE-3L's MRR changes by at most 0.010 under any perturbation, with all 24 comparisons equivalent to no change. However, this does not demonstrate robustness to shuffling: its original MRR is already very low (0.004 on SUSHI and 0.030 on TRUCE), leaving little performance to lose.

4.3.1 Limitations of D2

Order-invariant descriptions are rare. They make up 2.9% of SUSHI test descriptions, 3.3% of all classified TRUCE descriptions (245 of 7,334) and none of NOAA's. Natural description datasets are therefore dominated by order claims, which makes any DiD much less precise than the dependent-group change (e.g. for CLaSP at checkpoint 42 the 95% interval of the change has width about 0.04 over 715 dependent descriptions but about 0.20 over 18 invariant ones).

What the DiD isolates. It isolates less than its design suggests. Order-invariant descriptions mostly describe flat or constant series (the four SUSHI constant-shape descriptions, "The majority is flat."), which shuffling barely changes. A small invariant-group change is then expected from any model, so the DiD also reflects how much shuffling alters each series. Shuffled series are moreover unlike anything seen in training, which may explain why

TRACE loses 87–97% even under first-half shuffling. Time reversal and phase-randomised surrogates (Theiler et al., 1992), which change order but keep a series smooth, would separate the use of order from sensitivity to out-of-distribution inputs.

Perturbation strength. Masking replaces one fifth of the points; shuffling moves all of them. A larger loss under shuffling therefore does not show that order matters more than the values themselves.

4.3.2 Answer to RQ2

Full shuffling preserves all values but removes most matching performance on order-dependent descriptions: CLaSP loses 66–78% of MRR, TRACE loses 96–98%, and ChatTS falls to chance-level accuracy. CLaSP and TRACE retain some above-chance performance, examined in Section 4.4. Whether these losses are specific to descriptions that make claims about order can be assessed meaningfully only on TRUCE, since SUSHI has too few order-invariant descriptions and NOAA has none. On TRUCE, CLaSP shows an order-specific loss at two of three checkpoints, while ChatTS's large DiD mainly reflects an improvement in the order-invariant group. Similarly, TE-3L produces a DiD comparable to CLaSP's despite weak matching ability, driven by an improvement in its order-invariant group. Thus, DiD must be interpreted alongside the changes in both groups (Table 4.5 and Section E.3).

4.4 D3: What information supports matching after shuffling?

Figure 4.3 and Table 4.6 follow each model down the ladder. Higher scores indicate better matching performance (MRR for retrieval models and accuracy for ChatTS). Reading a row from left to right, the step at which the score falls to a reference level shows which information had been supporting it.

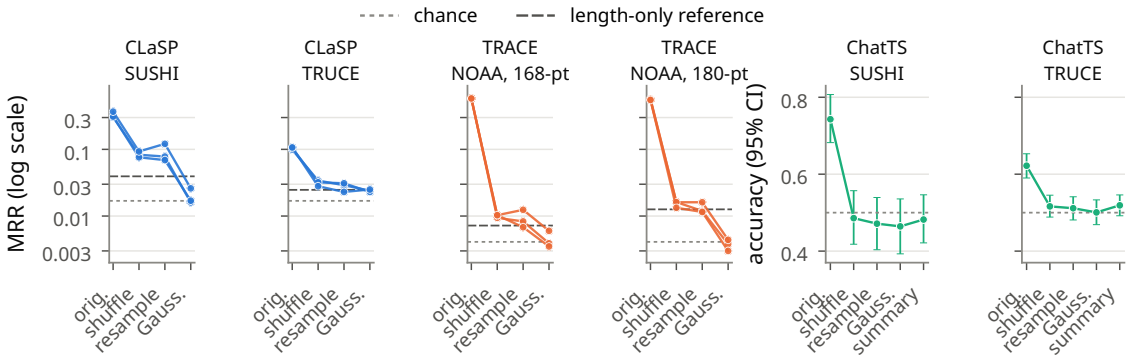


Figure 4.3: **D3 ladders.** Retrieval models (MRR, log scale; one line per run) and ChatTS (accuracy, 95% intervals). Dotted: chance; dashed: length-only reference. CLaSP on SUSHI stays above both references until the Gaussian step; CLaSP on TRUCE sits on the length-only reference from the shuffle step onwards; TRACE falls below the length-only reference only at the Gaussian step; ChatTS is at chance under every reduced input.

CLaSP on SUSHI. All SUSHI series have 2,048 points, so length cannot identify a SUSHI target among other SUSHI targets. The matched Gaussian brings CLaSP to chance (all intervals

Table 4.6: **D3 ladder.** Top: description→series MRR over all queries (CLaSP) or order-dependent queries (TRACE), ranges over three runs. Verdicts are all SHORTCUT PRESENT, naming the supporting information: the shape of the value distribution, series length, or distribution shape and/or cross-channel co-occurrence. Bottom: ChatTS accuracy, one run, 95% intervals; verdict NO SHORTCUT DETECTED on both datasets. Compare within a row only. Per-run values: Chapter F.

Model on data	Original	Shuffle	Resample	Gaussian	Chance	Len. ref.	Resample − Gaussian	Verdict
CLaSP, SUSHI	.304–.368	.076–.093	.069–.121	.016–.026	.017	.039	$+.052$ to $+.094$; all CIs > 0	shape
CLaSP, TRUCE	.099–.107	.028–.034	.023–.031	.023–.025	.017	.025	$-.002$ to $+.008$; all CIs $\ni 0$	length
TRACE, 168-pt	.579–.582	.0094–.0103	.0068–.0124	.0035–.0060	.0041	.0072	> 0 in every draw (pooled, $p_{\text{Holm}} \leq 3.4\times10^{-23}$)	shape and/or co-occurrence
TRACE, 180-pt	.551–.572	.0133–.0162	.0115–.0161	.0030–.0044	.0041	.0126		
TRACE, other	.478–.506	.0122–.0187	.0127–.0137	.0079–.0134	.0041	—		

Model on data	Original	Shuffle	Resample	Gaussian	Five-field summary
ChatTS, SUSHI	.743 [.682, .807]	.486 [.418, .557]	.471 [.404, .539]	.464 [.393, .536]	.482 [.421, .546]
ChatTS, TRUCE	.622 [.590, .653]	.516 [.489, .545]	.512 [.481, .541]	.501 [.469, .533]	.519 [.492, .546]

include 0.017), whereas own-value resampling keeps performance at or above the shuffled level and above the length-only reference (0.039). Resampling beats the Gaussian at every checkpoint: $+0.062\,[+0.033, +0.096]$, $+0.052\,[+0.020, +0.085]$ and $+0.094\,[+0.052, +0.141]$. Since standardisation removes mean and scale in both conditions and length is unchanged, the information that resampling keeps and the Gaussian removes is the *shape* of the value distribution, meaning which values are common and which are rare. Signals with spikes do not retain clearly more than clean or smooth ones (no interval excludes zero; Chapter F), so which distributional property matters remains open.

CLaSP on TRUCE. A 12-point TRUCE target competes with 2,048-point SUSHI series, so length alone narrows it to the 246 TRUCE series (length-only reference 0.025). The Gaussian, which leaves CLaSP essentially only length, matches this reference at every checkpoint (intervals contain 0.025), and resampling adds nothing measurable (resample−Gaussian intervals include zero). The residual is consistent with what length alone provides. It is an artefact of evaluating on a merged pool: in a per-dataset pool, where all candidates share one length, the length-only reference equals chance.

TRACE on NOAA. TRACE's residual is small (0.009–0.019). Resampling and shuffling remove equivalent shares of it (TOST equivalent in every draw), so exact values add nothing beyond their distribution. The Gaussian removes most of what resampling keeps: over all 2,005 order-dependent queries, resampling beats the Gaussian in every draw (Holm-adjusted $p \leq 3.4 \times 10^{-23}$). Length does not explain the residual: in the two large length groups, all six Gaussian estimates fall below their length-only references and four of six below chance. Because TRACE's resampling keeps whole time steps while its Gaussian draws channels independently, the design cannot tell whether TRACE uses per-channel distribution shape, cross-channel co-occurrence, or both. The pattern is not identical in every group and draw (in draw 14 the two large groups differ slightly under resampling, and the mixed-length group has a slightly higher Gaussian than resampled score), so the verdict applies to the order-dependent queries as a whole.

ChatTS. Under every reduced input, accuracy lies between 0.46 and 0.52 and every interval includes chance. Even the five-field summary (mean, standard deviation, minimum, maximum and length as text) stays at chance (0.482 and 0.519), far below the original. Re-pairing series and prompt numbers shows that ChatTS follows the series, not the statistics in its

prompt: a Gaussian series with the original series' numbers stays at chance, and the original series with *another* series' numbers is equivalent to the original (Chapter F). Therefore our verdict is NO SHORTCUT DETECTED under the tested reductions. The TRUCE result is limited by ChatTS's weak capability there and by TRUCE series being shorter than one 16-point patch.

Negative control. Resampling and the Gaussian leave TE-3L's order-dependent MRR within ± 0.05 in all 12 checks (largest change 0.0067), but, as in D2, its original MRR is too low for this to show robustness.

4.4.1 Robustness and limitations of D3

CLaSP's shape result on SUSHI holds against the length-only reference. CLaSP's Gaussian on SUSHI (0.016–0.026) falls below the length-only reference (0.039), so CLaSP does not fully use length in this condition. This could make the resample–Gaussian difference look larger than it is. Resampling, however, also lies above the length-only reference (0.069–0.121), so the distribution shape carries information beyond length either way.

TRACE's result is robust to padding and constant channels. The Gaussian is meant to keep only the mean, standard deviation and length of each series. For TRACE, however, it also leaves two things unchanged: the padding that fills each record up to 186 time steps, and the channels that are constant within a record. This extra information could help TRACE identify a record. Resampling, however, leaves the padding and constant channels unchanged as well, so they are present in both conditions and cannot explain why resampling scores higher than the Gaussian. That advantage must come from what resampling keeps and the Gaussian removes: the shape of the value distribution, or which values occur together across channels. Therefore, the D3 result for TRACE still holds: its residual rests on this information, not on padding or constant channels.

Whether exact values matter for CLaSP is unresolved. Four of the six resample-versus-shuffle tests for CLaSP are inconclusive. On TRUCE, resampling removes more than shuffling at all checkpoints, clearly only at checkpoint 44. One possible reason, not tested here, is that resampling only 12 values with replacement itself changes the distribution.

4.4.2 Answer to RQ3

The performance that remains after shuffling rests on different information in different settings: the shape of the value distribution for CLaSP on SUSHI, distribution shape and/or cross-channel co-occurrence for TRACE on NOAA, and series length for CLaSP on TRUCE, where length is informative only because the candidate pool mixes two datasets. The residual therefore has to be interpreted together with the candidate pool in which it is measured. Exact values add nothing beyond their distribution for TRACE. For CLaSP, whether they do remains unresolved. ChatTS retains nothing after shuffling, and explicit summary statistics do not restore its performance.

Chapter 5

Discussion and Conclusion

5.1 What the three diagnostics show together

The diagnostics are most informative in combination. D1 shows that high overall matching scores can hide failures on particular components, as also found in vision–language retrieval (Yuksekgonul et al., 2022). D2 shows that shuffling temporal order removes most matching performance on order-dependent descriptions. D3 examines what supports the performance that remains. For CLaSP on SUSHI, the residual depends on the distribution of values. On TRUCE, it is largely explained by series length because the candidate pool mixes 12-point TRUCE and 2,048-point SUSHI series. Thus, findings depend on the model, dataset, and candidate pool rather than the model alone.

Our results also relate to prior work in two ways. First, hard-negative training does not eliminate all component weaknesses. TRACE, which was trained with hard negatives, shows no COLLAPSE on the tested components but still loses accuracy on temperature-trend direction. Because TRACE was evaluated on different items from CLaSP and ChatTS, and also differs from them in architecture and training data, we cannot determine whether hard-negative training prevented collapse. Second, ChatTS does not use the numbers, namely the mean and scale of each series, in its prompt as a shortcut instead of reading the series. Changing the two numbers while keeping the series does not meaningfully change its answers. This does not contradict the strong prompt-prefix effect reported by Yin et al. (2026): their study tested adding or removing a richer prompt prefix in a smaller ChatTS-style model trained by the authors, whereas ours tested changing two prompt values while keeping the series unchanged.

5.2 Recommendations for evaluating alignment models

Our checks changed or qualified a conclusion in every diagnostic. The lessons below apply beyond the models studied here.

R1 Check that a metric can show failure before interpreting changes in it. Test judge-based and similarity-thresholded metrics on an untrained model first: if the untrained model also scores near-perfectly, the metric cannot reveal a drop (Table 4.2).

R2 Check that the model can do the task. A model that cannot do the task can still show statistically significant effects that mean nothing (TE-3L in D1), and "no change" means little when the original score is already below the equivalence margin (TE-3L in D2 and D3).

R3 Report composite statistics with their parts. Give a DiD with both group changes and a pooled effect with its subgroups. Otherwise an invariant-group *improvement* can look like order sensitivity (TE-3L, ChatTS on TRUCE), and pooling can hide subgroups that behave

in opposite ways (TRACE length groups).

R4 **Check what a condition actually removes, for each model and pool.** For CLaSP and TRACE, which standardise their input, the matched Gaussian leaves little beyond length, ChatTS still reads the mean and scale in its prompt. Length, in turn, helps only in pools that mix series lengths. Each such source needs its own reference or control: the length-only reference, and controls that change only the prompt numbers.

R5 **Review items by hand and check for text-only cues.** Human review changed swap accuracy in both directions (Section D.1). A swap can also make a multi-field description internally inconsistent, letting a model reject it without the series (Hsieh et al., 2023). Scoring both descriptions against unrelated series tests this.

R6 **Include descriptions that make no claim about order.** They are at most 3.3% of the descriptions studied, too few to test reliably whether a loss is specific to order.

5.3 Limitations

Behavioural, not mechanistic. The diagnostics show what a model's scores depend on under the tested changes. They do not reveal how the model works inside, and passing a diagnostic does not show general understanding.

Models and test format. Our results apply to the model versions and test format we used: a CLaSP *reimplementation* with a Transformer encoder instead of the original Informer, the released TRACE checkpoint, and ChatTS-14B choosing between two descriptions rather than writing free text. They may not carry over to the original CLaSP or to ChatTS's free-form answers. The models were also tested partly on different items and with different metrics, and their scores on the unchanged input differ widely. The results therefore show how each model behaves, but should not be used to judge which model is better.

Items and annotation. The component tests use templated (SUSHI) or generated (NOAA) descriptions rather than free-form text. All human checks were made by one annotator, so there is no inter-annotator agreement. Only fluctuation-type and temperature-trend swaps were checked one by one, while the other components were checked automatically, with small hand-checked samples for SUSHI only, and random-control descriptions were not checked by hand. TRACE could not be tested on fluctuation type, and D1 was not run on TRUCE.

Design choices made after seeing results. The ± 0.05 margin and the length-only reference were both fixed after some results had been seen (Section 3.8). Both were then applied unchanged to all later analyses.

Open explanations. Some results fit more than one explanation, and our design cannot separate them: TRACE's accuracy on location swaps, what supports its residual (distribution shape or cross-channel co-occurrence), and its length-group reversal in D2. These and the other limits of each diagnostic are discussed with the results (Sections 4.2 to 4.4). A shortcut is ruled out only for the reduced inputs we tested, other potential ones may remain.

5.4 Broader impact

Time-series–text models are being proposed for search and decision support over sensor, financial and environmental data. Our results suggest caution in reading their aggregate

scores as evidence that a description has been checked against a series: a system can retrieve a series for "temperature rose" while being much weaker at rejecting "temperature fell". The framework is a tool for exposing such gaps before deployment. It does not create new capabilities, and we see no direct route to misuse. The main risk is over-reading a passed diagnostic as a guarantee, which we caution against throughout.

5.5 Conclusion

A high matching score in time-series–text alignment does not explain itself. Across three datasets, the three alignment models studied rely mainly on the temporal order of the series, but they are not equally sensitive to every part of a description. Each model handles at least one description component poorly, a weakness its overall score does not show (RQ1). On descriptions that make claims about order, shuffling the series removes most of the above-chance performance (RQ2). What survives rests on different information in different settings, for example the distribution of values, or series length in a candidate pool that mixes series lengths (RQ3). Controlled diagnostics with built-in validity checks can show what lies behind such a score. Future work should separate the explanations left open here, test components on free-form descriptions with several annotators and text-only controls, use order perturbations that keep the series smooth, and add analyses of internal representations.

Declaration on the use of generative AI

During this thesis, I used the generative AI assistant Claude (Anthropic) to support writing, the preparation of figures and tables, and coding. For writing, Claude helped improve the academic style, clarity and structure of the thesis. For figures and tables, it helped adjust plotting code and table layouts to present the results more clearly. For coding, it helped debug and modify experiment and analysis scripts. I take full responsibility for all content in this thesis, including the text, figures, tables and code produced with AI assistance.

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Appendix A

CLaSP Reimplementation

No official code was available, so CLaSP was reimplemented from the paper (Ito et al., 2025). Table A.1 lists the settings: where the paper specifies a setting we follow it, and where it is silent we state our choice. The main deviation is the signal encoder. The paper uses an Informer (Zhou et al., 2021), whose distinctive mechanisms approximate full attention to reduce its cost on long series; we use a standard Transformer encoder with exact attention, which is tractable at the maximum length of 2,048 points.

Table A.1: **Settings of the CLaSP reimplementation** compared with the paper.

Setting	Paper	This study
Signal encoder	Informer, trained from scratch	Standard Transformer encoder, exact attention, trained from scratch; 3 layers, width 512, 8 heads
Text encoder	T5-Small, trained jointly	T5-Small (`google-t5/t5-small`), fine-tuned jointly
Common space, similarity	Linear projections; temperature-scaled cosine similarity	Dimension 512; learnable temperature, initialised at $1/0.07$
Loss	Symmetric cross-entropy over in-batch pairs	Same
Data and splits	One joint model on TRUCE and SUSHI; 8:1:1	Same; SUSHI Tiny (public, 1,400 series), class-stratified split with seed 42
Pooling	Not specified	Mean over valid positions, for series and text
Series normalisation	Not specified	Per-series standardisation to mean 0, s.d. 1
Batches	Not specified	One dataset per batch: 64 (TRUCE), 8 (SUSHI)
Optimiser	Not specified	AdamW; lr 10^{-4} (new parameters), 10^{-5} (T5); weight decay 0.01
Stopping	Not specified	Up to 100 epochs; early stopping on validation loss, patience 10
Training seeds	Not specified	42, 43, 44; stopped after 21, 26 and 24 epochs

A.1 Comparison with the published values

Two differences in setting prevent a direct comparison of Table 4.2 with the published values. The first is the candidate pool. We rank every description against a merged pool of 386 test series (140 SUSHI, 246 TRUCE). The paper does not state the pool behind each value, but its combined row cannot be a query-weighted average of its two per-dataset rows: the share of TRUCE queries that would produce it is 5.3%, 3.4%, 6.4% and 2.2% in the four configurations, which no single query set can satisfy. The per-dataset values were therefore most likely computed against smaller, dataset-specific pools, which make retrieval easier.

The second is the SUSHI version. The public Tiny release (1,400 series) leaves one test series per class under the 8:1:1 split. The paper does not name its version, but its SUSHI values are consistent with the larger Base version, which would leave about one hundred test series per class with near-identical template descriptions. Because the judge-based metric credits any retrieved series whose description is similar enough to the query, a query in the Base setting has many series that count as relevant, whereas a query in the Tiny setting has far fewer. Consistently, our shortfall on SUSHI grows as the judge becomes stricter: with Sentence-BERT it is 0.128 at threshold 0.5 (0.854 vs. 0.982) and 0.265 at threshold 0.8 (0.306 vs. 0.571).

Appendix B

Data, Item Construction and Human Validation

B.1 Data examples

Figure B.1 shows one SUSHI series and two TRUCE series with their captions; Figure B.2 and Table B.1 show one NOAA record (Mariposa, California, July–December 2022; 180 time steps per variable) and its ten description fields.

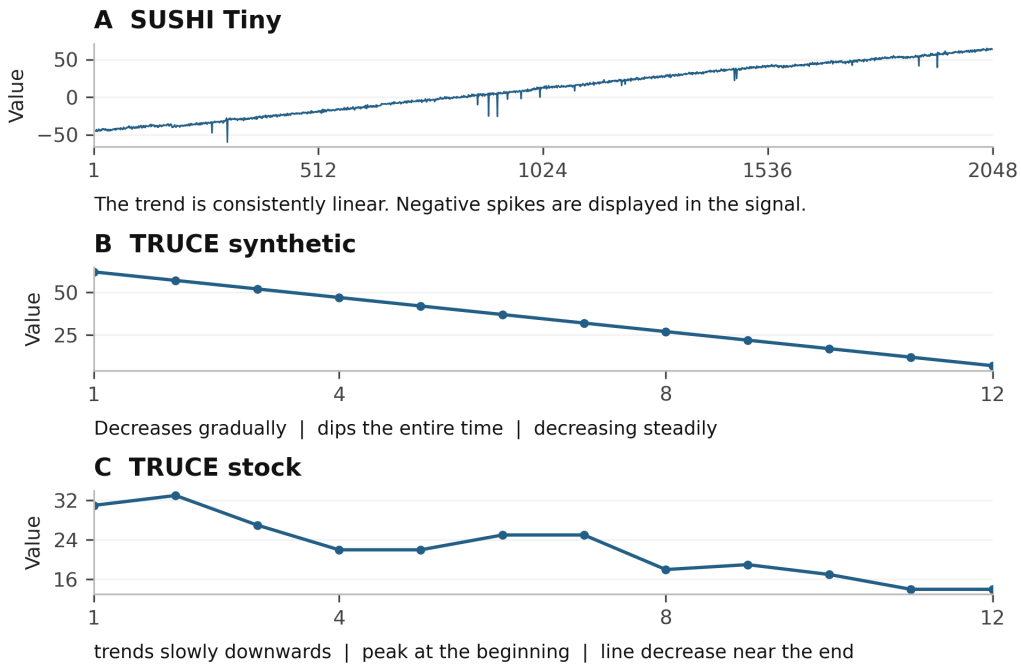


Figure B.1: **SUSHI and TRUCE examples.** Horizontal axes count time steps. SUSHI has one caption per series; all three captions are shown for each TRUCE series.

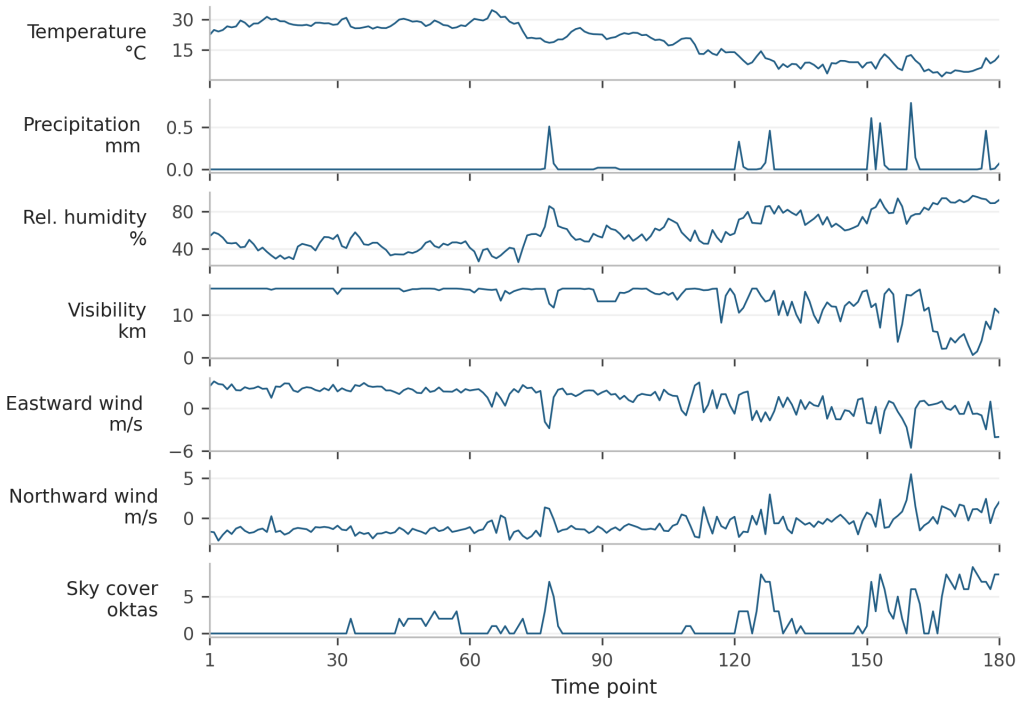


Figure B.2: **NOAA example** (test record 206). The seven variables share one time axis. Values are shown before the model's normalisation and padding.

Table B.1: **The ten text fields of NOAA record 206.** Long fields are shortened (…).

Field	Stored text
DATE (time span)	The past six months from July 2022 to December 2022.
`location`	Mariposa, California experienced a diverse range of weather conditions.
`labels`	Clear, Hot, Cold, Cloudy, Humid
`temperature`	The temperature showed a noticeable trend of rising during the summer months, peaking in August with highs around 34.6 °C, followed by a gradual decline into colder temperatures in December…
`precipitation`	Precipitation was minimal, with most days recording no rainfall, but there were sporadic instances of light rain…
`relative_humidity`	Relative humidity fluctuated throughout the period, generally decreasing during the warmer months…
`visibility`	Visibility remained mostly stable, averaging around 16.09 km, with occasional drops…
`wind_u` (eastward)	Wind direction also showed variability, with occasional strong gusts recorded, particularly in late October.
`wind_v` (northward)	Wind velocity varied, with some days experiencing calm conditions and others showing gusts…
`sky_code` (cloud cover)	The sky cover varied from clear to overcast…

B.2 Construction and screening of the D1 items

SUSHI. Items were built from the 279 validation and test series. Each swap item is paired with a random-control item on the same series, giving 2,770 swap and 2,770 random-control items: 205 for C1, 410 for C2, 280 for C3, 990 for C4 and 885 for C5, in each condition. Automated checks on all 2,770 swap items confirmed that every wrong description equals the correct one with exactly one recorded sentence replaced, that the replaced sentence occurs in the original, and that no replacement leaves the text unchanged. For C4, 107 of 990 swaps were removed because the new sentence did not explicitly name the fluctuation type (e.g. spikes without a direction), and 20 were set aside because the swapped word could also describe the shape (e.g. step changes claimed of a sawtooth, whose resets are abrupt jumps). The remaining 863 were checked by hand (Section B.3); 125 failed, leaving 738 certified swaps.

NOAA. Items were generated by rule for each component, with 400 swap and 400 random-control items per component (seed 42). Records whose swapped description would be inconsistent were removed before sampling. For N1, the swap must not produce a contradictory label set such as Dry with Humid; this removed 199 of 661 otherwise eligible records, leaving 462. For N3, descriptions mentioning both rising and falling temperature were excluded, among other rules, leaving 777 eligible records. N2 swaps one week with six months, so 28-day records are not used. After generation, an automatic search found 32 N3 items with wording about temperature peaks, 15 of which tie the peak to a particular time; 11 of these 15 were judged defective and removed with their random controls, leaving 389 N3 swaps for the human check.

Why TRACE was not tested on fluctuation type. First, SUSHI series could not be used. TRACE receives at most 186 points, and when the 2,048-point SUSHI series were shortened to this length, the signal classifier of Section D.1, applied to all 1,400 SUSHI series, could no longer reliably separate fluctuation types: mean accuracy over fluctuation pairs fell from 0.929 to 0.773, and three spike-related pairs to 0.525–0.595, close to chance, while global-shape components stayed at 0.93–0.99. Interpolation, decimation and fixed windows gave the same result (0.773, 0.784, 0.786). Second, a NOAA fluctuation component (N4, fluctuation vs. stability) could be built for only 33 of 2,006 records, because NOAA descriptions of fluctuation almost always cite supporting values or ranges that a one-word edit cannot change consistently. N4 was dropped; the other components keep their original numbers.

B.3 Human validation

All human checks were made by the author. A swap counts as valid only if its wrong description is clearly false for the signal to a competent reader, without relying on the dataset's own conventions for its class words. Each item was judged on three questions: is the edited sentence well formed, does it actually state the new value, and is that statement false for the signal? Samples were used to find recurring defects; C4 on SUSHI and N3 on NOAA were then checked item by item, and D1 results for these two components use certified items only. Random-control descriptions were not checked by hand.

SUSHI. A stratified sample of 50 swaps (10 per component) gave 46 valid items. Two failures were fluctuation sentences that state no direction or size and remain true of the signal; one was a step-change claim about a sawtooth signal; one was a hedged trend-family sentence ("akin to a sigmoid") that fits the true shape. Under a reading based on the dataset's own class vocabulary, two of the four would pass (48/50); the stricter plain-language reading

was applied throughout. All 863 remaining C4 swaps were read in one fixed random order in nine batches; every failure belongs to one of five mechanisms (Table B.2), and repeated sentences received the same verdict every time.

Table B.2: **Why SUSHI fluctuation-type swaps failed the human check.**

Mechanism	What happens	Items
Subset claim	The signal has both positive and negative spikes; the new sentence claims spikes of one direction only, which is literally true	66
Non-pervasive noise	Spike signal; the new sentence claims noise without saying it is pervasive, which spikes can satisfy	42
Bare noise statement	Step signal; the new sentence states that noise is present without any size, which every SUSHI signal satisfies	3
Truncated sentence	The new sentence comes from a malformed SUSHI description beginning “Large part,”	10
Reverse overlap	Noisy signal; the new sentence claims frequent positive and negative spikes without any size, which describes noise itself	4
Total		125

NOAA. The 389 N3 swaps were checked in two stages: a stratified sample of 100 (17 defects), then the remaining 289 (28 defects), so every N3 swap was judged by hand. Defects kept the original direction elsewhere in the description, for example an appended interpretation such as “indicating a cold snap”, a humidity sentence stating how temperature changed, or named months in a six-month span that imply the season’s direction. The 45 defective swaps were removed with their random controls, leaving 344.

B.4 Order grouping for D2

A description is order-dependent if shuffling the series would make any of its claims false, order-invariant if it makes no claim that depends on order, and unclear otherwise. Rules applied this grouping automatically, and the result was checked by hand. **SUSHI**: all 40 descriptions in the order-invariant classes (constant shapes with a fluctuation) were confirmed; four are in the test set. **TRUCE**: all 189 distinct descriptions placed in the order-invariant group and all 43 in the unclear group were checked, together with a sample of 30 order-dependent descriptions (all confirmed). **NOAA**: a sample of 17 descriptions confirmed the rules’ output (16 order-dependent, one unclear).

The hand checks changed the rules’ grouping in a few cases. The SUSHI description “Trend unchanged, not alternating from value” was kept as order-invariant: its order word is negated, so the claim survives shuffling. In TRUCE, eight descriptions changed group, all from the training or validation split, so test-set counts are unaffected. Five moved from order-invariant to order-dependent; e.g. “the changes stay almost constant throughout the pattern” can describe a steady rise, which shuffling does not preserve, and “stay flat in the fist part.” refers to a position despite its misspelling. Of the unclear descriptions, “highly placed line” was moved to order-invariant because it states only the level, and two misspelled descriptions of how the series rises or falls (e.g. “incresae the same”) to order-dependent. Each sampled order-dependent description, in TRUCE and NOAA, ties a direction, shape or value to a position or period (e.g. “increases at the end.”; “a general warming trend observed towards the end of the period”). The single unclear NOAA description belongs to a record whose temperature, precipitation and relative humidity are constant throughout the week; for these variables, as for the constant SUSHI signal, shuffling changes nothing.

Appendix C

Construction of the Diagnostic Conditions

C.1 Rules shared by all conditions

All conditions change only the series; descriptions, model weights and candidate pools stay the same. For CLaSP, TRACE and TE-3L, every random draw is seeded separately for each series from the series identifier, the condition and the run number, so each condition is exactly reproducible and runs differ only in their run number. CLaSP and TE-3L receive identical perturbed series in runs 42, 43 and 44. TRACE uses runs 13, 14 and 15, which also determine its built-in evaluation mask.

C.2 D2

Figure C.1: **What each D2 condition changes**, illustrated with a ten-point series. Letters are observations, the slash marks the middle, and m is a masked point that the model receives as zero. In the experiments, masking affects $\lfloor L/5 \rfloor$ points.

Condition	Example
Original	a b c d e / f g h i j
Full shuffle	h c a j f / b i e g d
First-half shuffle	c a e b d / f g h i j
Half exchange	f g h i j / a b c d e
Masking	a m c d e / f g m i j

The three shuffles follow Tan et al. (2024). Full shuffling applies a random permutation to all points; first-half shuffling permutes only the first half; half exchange swaps the two halves and keeps the order within each. Each half has 1,024 points for SUSHI and 6 for TRUCE. For TRACE, halves are formed from valid points only and padding stays in place; all seven channels receive the same permutation, so values recorded at one time step stay together.

Masking selects $\lfloor 0.2L \rfloor$ of the L valid points at random and sets them to zero as the model receives them: 409 of 2,048 points for SUSHI and 2 of 12 (16.7%) for TRUCE. For TRACE, rounding down per series gives an effective masked share of 0.1976. For CLaSP, TRACE and TE-3L, masked points are set to zero after standardisation. For ChatTS, they are first filled with the mean of the remaining points, which becomes exactly zero once ChatTS subtracts

the series mean. For TRACE, the D2 change is applied before TRACE's own 30% evaluation mask, which is identical across conditions; the overlap between the two masks was not measured, and the effective share of 0.1976 does not depend on it.

ChatTS processes a 2,048-point SUSHI series as 128 patches of 16 points. Within-patch shuffling permutes the 16 values inside each patch and keeps the patch order; patch-order shuffling permutes the patches and keeps each patch's internal order. A 12-point TRUCE series is shorter than one patch, so these two conditions are SUSHI-only.

C.3 D3

Reduced series are built from raw values and then pass through each model's usual preprocessing. The shuffled condition reuses the D2 full shuffle. Resampling draws L values with replacement from the series' own values; the matched Gaussian draws L values from $\mathcal{N}(\mu, \sigma^2)$ with the series' mean and standard deviation. All keep the original length.

For TRACE, reduced series are built from the valid points of each channel and padding is unchanged. Resampling draws time steps with replacement and copies all seven channel values at each drawn step, so co-occurring values stay together; the matched Gaussian draws each channel independently. Channels that are constant in the original (498 in the test set) pass through unchanged. If resampling turned a channel with few distinct values into a constant one, the time steps were redrawn, up to eight times.

For models that standardise, matching mean and standard deviation adds no information: standardising $\mu + \sigma z$ gives exactly the standardised z. For TRACE this was also checked numerically (maximum difference 2.4×10^{-7}), as was building a condition before or after TRACE's per-channel standardisation (maximum difference 9.5×10^{-7}).

The length-only reference (Section 3.2.3) uses $n = 140$ for a SUSHI target and $n = 246$ for a TRUCE target in the CLaSP pool, giving 0.039 and 0.025; with $n = 386$ the same formula gives the chance level 0.017. For TRACE, n is the number of candidates with the target's unpadded length.

C.4 Additional conditions for ChatTS

ChatTS centres each series, rescales it only when its values are large, and states the removed offset and scale as numbers in its prompt, so a change to the series can also change these numbers. Three conditions separate the two effects. (1) The *prompt-number control* accompanies D2 masking: 100 descriptions per dataset are asked in both orders with the series unchanged, while the prompt numbers are shifted by exactly the amount masking produces for that series. (2) Condition A gives the matched Gaussian series with the *original* series' prompt numbers. (3) Condition B gives the original series with the prompt numbers of a different, randomly chosen series from the same dataset (never the description's own). Replacing rather than deleting the numbers keeps the prompt format unchanged. Condition C gives a Gaussian series with its own numbers, built through the same route, as a check on the construction. The five-field summary replaces the series entirely with one line stating mean, population standard deviation, minimum, maximum and length, each to four decimals.

Because ChatTS's standard input processor cannot pair a series with numbers that do not belong to it, these prompts were built by a separate routine reusing the arithmetic of ChatTS's processing code. On all 1,756 D2 inputs (878 descriptions $\times$ 2 answer orders) it reproduces the standard processor bit for bit, and condition C gives the same answer as the standard Gaussian condition on every question on both datasets.

Appendix D

Supplementary Results for D1

D.1 Further checks

Verdicts without the item review. Fluctuation type and temperature-trend direction are evaluated only on swaps that passed an item-by-item human review. Table D.1 compares these results with those on all swaps. The review changes the numbers but not the verdicts, and the changes go in different directions: swap accuracy changes little for CLaSP, rises for ChatTS and falls for TRACE, because the 45 swaps removed for TRACE were ones it found relatively easy (0.800–0.844). The review therefore does not systematically make the models look better or worse.

Table D.1: **D1 verdicts with and without the item review.** Swap accuracy on all swaps and on the swaps that passed the item-by-item human review; n in brackets. Ranges span three CLaSP checkpoints or three TRACE mask draws; ChatTS has one run.

		All swaps		Certified swaps	
Model on data	Component	Swap acc.	Verdict	Swap acc.	Verdict
CLaSP on SUSHI	fluctuation type	.593–.603 (990)	COLLAPSE	.592–.618 (738)	COLLAPSE
ChatTS on SUSHI	fluctuation type	.658 (990)	PARTIAL	.690 (738)	PARTIAL
TRACE on NOAA	temperature-trend direction	.720–.725 (389)	PARTIAL	.703–.712 (344)	PARTIAL

The location cannot reach TRACE's series embedding. TRACE's retrieval score is the cosine similarity between a series embedding and a description embedding. In TRACE's code, the series embedding is computed from the seven weather series before the cross-attention step; that step, which uses channel-level text, changes only channel-level outputs that the retrieval score does not use. The location text therefore cannot reach the series side of the score at test time.

Signal classifier. The classifier uses 16 measurements of the standardised signal. Six describe global shape: linear, quadratic and cubic fit terms, the difference between the first and last thirds, the difference between the middle and the ends, and residual skew. Ten describe local texture: step volatility, kurtosis and skew of the differences, the largest normalised jump, spike rate, lag-1 autocorrelation, residual roughness and kurtosis, size of level shifts, and the rate of direction reversals. For each of the 124 value pairs within a component, a logistic regression on the standardised measurements separates the two values, scored by stratified 5-fold cross-validation (seed 42) on the 279 D1 series; a component's accuracy is the mean over its pairs.

Description-length checks. The first check applies two length-only rules (always choose the longer description, or always choose the shorter; ties are counted as half correct) to every component and condition. The second, for CLaSP and TRACE, correlates the word-count difference between the two descriptions with the model's score difference across swap items. ChatTS gives only a choice, so only the first check applies. For N1 and N3, both descriptions of every swap have the same number of words.

D.2 Per-run results and tests

Intervals are 95% bootstrap intervals over signals (10,000 resamples). The paired bootstrap test is two-sided and Holm-adjusted within each run across components; a bootstrap p of zero (no resample crossed zero) corresponds after adjustment to $p < 0.001$. The Wilcoxon signed-rank test on per-signal accuracy differences is unadjusted. For CLaSP C4, the certified-item intervals come from a separate run of the same procedure, and their p-values replace the all-item p-value in each checkpoint's Holm adjustment.

Table D.2: **D1 for CLaSP on SUSHI**, per checkpoint. † Certified swaps.

Component	Ckpt	Random	Swap [95% CI]	Gap [95% CI]	Boot. p (Holm)	Wilcoxon p
C1 trend direction	42	0.990	0.898 [0.854, 0.937]	+0.093 [+0.049, +0.137]	< 0.001	7.4×10^{-5}
	43	0.980	0.907 [0.868, 0.941]	+0.073 [+0.034, +0.112]	< 0.001	5.8×10^{-4}
	44	0.985	0.927 [0.893, 0.961]	+0.059 [+0.024, +0.093]	0.0006	0.0013
C2 trend family	42	0.978	0.934 [0.902, 0.963]	+0.044 [+0.017, +0.073]	0.0010	0.010
	43	0.990	0.956 [0.927, 0.980]	+0.034 [+0.010, +0.062]	0.0062	0.022
	44	0.993	0.963 [0.934, 0.988]	+0.029 [+0.007, +0.056]	0.0136	0.0094
C3 periodic waveform	42	0.925	0.750 [0.689, 0.811]	+0.175 [+0.111, +0.239]	< 0.001	7.7×10^{-6}
	43	0.932	0.746 [0.686, 0.804]	+0.186 [+0.132, +0.243]	< 0.001	3.3×10^{-7}
	44	0.918	0.732 [0.668, 0.793]	+0.186 [+0.129, +0.246]	< 0.001	5.8×10^{-7}
C4 fluctuation type†	42	0.970	0.600 [0.551, 0.650]	+0.370 [+0.319, +0.421]	< 0.001	4.1×10^{-25}
	43	0.970	0.592 [0.545, 0.640]	+0.378 [+0.329, +0.427]	< 0.001	1.6×10^{-25}
	44	0.973	0.618 [0.570, 0.667]	+0.355 [+0.304, +0.405]	< 0.001	8.6×10^{-23}
C5 signal regime	42	0.951	0.989 [0.976, 0.999]	−0.037 [−0.053, −0.022]	< 0.001	1.6×10^{-4}
	43	0.949	0.980 [0.955, 0.996]	−0.031 [−0.045, −0.016]	< 0.001	0.0032
	44	0.957	0.983 [0.967, 0.996]	−0.026 [−0.045, −0.007]	0.0136	0.0091

Table D.3: **D1 for TRACE on NOAA**, per mask draw. † Certified swaps.

Component	Draw	Random	Swap [95% CI]	Gap [95% CI]	Boot. p (Holm)	Wilcoxon p
N1 weather labels	13	1.000	0.893 [0.860, 0.923]	+0.108 [+0.078, +0.140]	< 0.001	5.5×10^{-11}
	14	0.993	0.880 [0.848, 0.910]	+0.113 [+0.083, +0.145]	< 0.001	5.2×10^{-11}
	15	1.000	0.875 [0.843, 0.905]	+0.125 [+0.095, +0.157]	< 0.001	1.5×10^{-12}
N2 time span	13	0.990	0.938 [0.913, 0.960]	+0.052 [+0.027, +0.080]	< 0.001	9.6×10^{-5}
	14	0.998	0.935 [0.910, 0.958]	+0.063 [+0.040, +0.088]	< 0.001	1.5×10^{-6}
	15	1.000	0.950 [0.928, 0.970]	+0.050 [+0.030, +0.073]	< 0.001	7.7×10^{-6}
N3 temperature trend†	13	1.000	0.703 [0.654, 0.753]	+0.297 [+0.247, +0.346]	< 0.001	5.6×10^{-24}
	14	0.994	0.712 [0.666, 0.759]	+0.282 [+0.233, +0.331]	< 0.001	4.8×10^{-22}
	15	0.997	0.709 [0.663, 0.759]	+0.288 [+0.238, +0.337]	< 0.001	6.8×10^{-23}
N5 location	13	1.000	0.925 [0.898, 0.950]	+0.075 [+0.050, +0.103]	< 0.001	4.3×10^{-8}
	14	1.000	0.935 [0.910, 0.958]	+0.065 [+0.042, +0.090]	< 0.001	3.4×10^{-7}
	15	1.000	0.918 [0.890, 0.943]	+0.083 [+0.057, +0.110]	< 0.001	9.2×10^{-9}

Table D.4 gives the equivalence tests mentioned in Section 4.2 (four decimals so that intervals near the margin can be read). For CLaSP signal regime at checkpoint 42 the interval extends 0.0005 beyond the margin in the direction of higher swap accuracy, so the outcome is shifted. For ChatTS, intervals come from 2,000 resamples over signals. Table D.5 gives the

ChatTS tests: the main test is the Wilcoxon signed-rank test on per-signal differences, Holm-adjusted across the five components, with McNemar's test as a second check. Fluctuation type uses the certified swaps; its Holm-adjusted p is at most five times its raw value, hence below 0.001. Stored p-values have six decimals, so values stored as zero are reported as $< 10^{-6}$. Periodic waveform is VOID and its tests are not interpreted; tests for TE-3L are not reported because every TE-3L component is VOID.

Table D.4: **Equivalence tests for the D1 gap** (± 0.05, 90% intervals).

Model on data	Component	Run	Gap	90% CI	Outcome
CLaSP on SUSHI	C5 signal regime	ckpt 42	-0.0373	$[-0.0505, -0.0249]$	shifted
		ckpt 43	-0.0305	$[-0.0422, -0.0184]$	equivalent
		ckpt 44	-0.0260	$[-0.0419, -0.0099]$	equivalent
ChatTS on SUSHI	C1 trend direction	single	-0.0366	$[-0.0878, +0.0122]$	inconclusive
	C5 signal regime	single	-0.0147	$[-0.0465, +0.0188]$	equivalent

Table D.5: **Paired tests for ChatTS on SUSHI in D1.** † Certified swaps.

Component	Wilcoxon p	Wilcoxon p (Holm)	McNemar p
C1 trend direction	0.213	0.561	0.545
C2 trend family	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$
C3 periodic waveform (VOID)	0.187	0.561	0.430
C4 fluctuation type†	2.2×10^{-5}	< 0.001	$< 10^{-6}$
C5 signal regime	0.342	0.561	0.233

Appendix E

Supplementary Results for D2

A change is the score under a perturbation minus the original score (negative = loss); the share lost is the loss divided by the original score; $\mathrm{DiD} = \Delta_{\mathrm{inv}} - \Delta_{\mathrm{dep}}$. Unless stated otherwise, intervals are 95% percentile bootstrap intervals over signals with 2,000 resamples, and equivalence tests use the 90% interval and the ± 0.05 margin, reported to four decimals.

E.1 Group counts

Table E.1: **Order-dependent and order-invariant descriptions.** The invariant share is over dependent plus invariant descriptions. In the TRUCE test set, the 715 dependent descriptions cover 245 series and the 18 invariant ones 14 series; in SUSHI and NOAA each description belongs to a different series.

Dataset	Descriptions	Dependent	Invariant	In neither group	Invariant share
SUSHI	test set	135	4	1 (constant signal)	2.9%
TRUCE	all classified	7,089	245	46 (unclear)	3.3%
TRUCE	test set	715	18	5 (unclear)	2.5%
NOAA	test set	2,005	0	1 (unclear)	0.0%

E.2 CLaSP

The Wilcoxon signed-rank test compares each description's reciprocal rank before and after the perturbation; for CLaSP it is one-sided (testing for a loss) and unadjusted.

Table E.3 gives the MRR remaining after full shuffling with its 95% interval (from the D3 bootstrap over the same ranks, 10,000 resamples). Chance is 0.0169, and every lower bound lies above it.

Table E.2: **D2 for CLaSP on order-dependent descriptions**, per checkpoint. † The 95% interval of the change includes zero.

Data	Condition	Ckpt	Orig. MRR	MRR after	Change [95% CI]	Share lost / Wilcoxon p
SUSHI	full shuffle	42	0.304	0.072	−0.232 [−0.289, −0.177]	0.764 / 3.8×10^{-17}
		43	0.309	0.070	−0.239 [−0.293, −0.185]	0.774 / 4.4×10^{-18}
		44	0.358	0.078	−0.280 [−0.343, −0.219]	0.783 / 1.1×10^{-16}
	first-half shuffle	42	0.304	0.259	−0.045† [−0.101, +0.011]	0.147 / 0.0082
		43	0.309	0.233	−0.076 [−0.132, −0.016]	0.247 / 7.0×10^{-5}
		44	0.358	0.294	−0.064 [−0.123, −0.010]	0.180 / 0.023
	half exchange	42	0.304	0.081	−0.222 [−0.273, −0.176]	0.733 / 1.8×10^{-17}
		43	0.309	0.074	−0.235 [−0.291, −0.182]	0.761 / 2.6×10^{-16}
		44	0.358	0.115	−0.243 [−0.312, −0.170]	0.678 / 1.1×10^{-12}
	masking	42	0.304	0.181	−0.123 [−0.167, −0.078]	0.405 / 4.0×10^{-10}
		43	0.309	0.160	−0.149 [−0.201, −0.098]	0.481 / 3.2×10^{-9}
		44	0.358	0.304	−0.054 [−0.103, −0.008]	0.151 / 0.0091
TRUCE	full shuffle	42	0.100	0.034	−0.066 [−0.085, −0.047]	0.657 / 1.6×10^{-43}
		43	0.108	0.032	−0.075 [−0.096, −0.057]	0.699 / 2.0×10^{-43}
		44	0.110	0.028	−0.082 [−0.100, −0.064]	0.745 / 7.2×10^{-45}
	first-half shuffle	42	0.100	0.080	−0.020 [−0.035, −0.007]	0.200 / 2.7×10^{-8}
		43	0.108	0.081	−0.027 [−0.044, −0.010]	0.248 / 1.5×10^{-5}
		44	0.110	0.092	−0.018 [−0.035, −0.002]	0.162 / 5.0×10^{-5}
	half exchange	42	0.100	0.020	−0.080 [−0.099, −0.062]	0.798 / 2.3×10^{-53}
		43	0.108	0.017	−0.090 [−0.111, −0.072]	0.840 / 3.8×10^{-58}
		44	0.110	0.022	−0.088 [−0.108, −0.070]	0.801 / 3.4×10^{-54}
	masking	42	0.100	0.086	−0.014† [−0.032, +0.004]	0.140 / 0.0075
		43	0.108	0.093	−0.015† [−0.031, +0.002]	0.137 / 5.5×10^{-4}
		44	0.110	0.108	−0.002† [−0.018, +0.015]	0.018 / 0.154

Table E.3: **MRR remaining after full shuffling for CLaSP**, order-dependent descriptions.

Data	ckpt 42	ckpt 43	ckpt 44
SUSHI	0.072 [0.043, 0.105]	0.070 [0.044, 0.102]	0.078 [0.052, 0.109]
TRUCE	0.034 [0.025, 0.047]	0.032 [0.026, 0.040]	0.028 [0.022, 0.035]

Table E.4: **Full-shuffle group comparison for CLaSP**, per checkpoint. The DiD interval resamples signals independently within each group. SUSHI rows are descriptive (four invariant descriptions).

Data	Ckpt	Dep. orig.	Δ_{dep} [95% CI]	Inv. orig.	Δ_{inv} [95% CI]	DiD [95% CI]
SUSHI	42	0.304	−0.232 [−0.289, −0.177]	0.375	+0.108 [−0.117, +0.358]	+0.340 [+0.107, +0.608]
	43	0.309	−0.239 [−0.293, −0.185]	0.431	−0.138 [−0.375, +0.006]	+0.102 [−0.142, +0.271]
	44	0.358	−0.280 [−0.343, −0.219]	0.667	−0.125 [−0.375, +0.083]	+0.155 [−0.108, +0.384]
TRUCE	42	0.100	−0.066 [−0.085, −0.047]	0.077	−0.064 [−0.199, −0.003]	+0.002 [−0.128, +0.072]
	43	0.108	−0.075 [−0.096, −0.057]	0.021	+0.001 [−0.014, +0.021]	+0.076 [+0.051, +0.106]
	44	0.110	−0.082 [−0.100, −0.064]	0.024	−0.009 [−0.020, +0.002]	+0.073 [+0.052, +0.094]

Table E.5: **Equivalence tests for the order-invariant TRUCE group of CLaSP** under full shuffling (18 descriptions, 14 series). At checkpoint 42 the interval excludes zero and exits the margin.

Checkpoint	Orig. MRR	Change	90% CI	Outcome
42	0.0769	−0.0635	[−0.1795, −0.0051]	shifted
43	0.0205	+0.0009	[−0.0122, +0.0162]	equivalent
44	0.0239	−0.0092	[−0.0182, +0.0003]	equivalent

E.3 TE-3L negative control

Each TE-3L draw uses the same random shuffles and masks as the CLaSP checkpoint with the same number. The largest absolute change is 0.0105 (TRUCE, full shuffle, draw 43). Because the original MRR is below the margin on both datasets, even a complete loss would count as equivalent.

On TRUCE, the DiDof TE-3L is positive in all three draws, with intervals excluding zero in draws 43 and 44, as for CLaSP (Table 4.5). The cause differs. At checkpoints 43 and 44, CLaSP's DiDcomes from a drop in the order-dependent group, whereas TE-3L's comes mainly from a rise in the order-invariant group. Removing the two invariant descriptions of one series reduces TE-3L's DiDby more than half in draws 43 and 44 (Table E.7).

Table E.6: **Equivalence tests for TE-3L on order-dependent descriptions** (all 24 equivalent). Original MRR: 0.0037 (SUSHI), 0.0295 (TRUCE).

Data	Condition	Draw 42: change [90% CI]	Draw 43	Draw 44
SUSHI	full shuffle	−0.0004 [−0.0007, −0.0001]	−0.0004 [−0.0008, −0.0002]	−0.0004 [−0.0007, −0.0002]
	first-half shuffle	+0.0017 [−0.0002, +0.0052]	+0.0017 [−0.0001, +0.0052]	+0.0024 [−0.0001, +0.0070]
	half exchange	+0.0008 [−0.0000, +0.0018]	+0.0008 [−0.0000, +0.0018]	+0.0008 [−0.0000, +0.0018]
	masking	−0.0001 [−0.0003, +0.0002]	−0.0001 [−0.0003, +0.0001]	−0.0001 [−0.0004, +0.0001]
TRUCE	full shuffle	−0.0096 [−0.0162, −0.0033]	−0.0105 [−0.0169, −0.0045]	−0.0073 [−0.0138, −0.0012]
	first-half shuffle	−0.0038 [−0.0106, +0.0024]	+0.0004 [−0.0060, +0.0070]	−0.0010 [−0.0072, +0.0060]
	half exchange	−0.0096 [−0.0164, −0.0030]	−0.0096 [−0.0169, −0.0024]	−0.0096 [−0.0165, −0.0028]
	masking	−0.0003 [−0.0068, +0.0066]	−0.0025 [−0.0079, +0.0032]	+0.0008 [−0.0060, +0.0076]

Table E.7: **Full-shuffle group comparison for TE-3L on TRUCE.** The last column removes the two invariant descriptions of one series (16 remain); no interval was computed for it.

Draw	Dep. orig.	Δ_{dep} [95% CI]	Inv. orig.	Δ_{inv} [95% CI]	DiD [95% CI]	DiD w/o one series
42	0.030	−0.010 [−0.018, −0.002]	0.071	+0.001 [−0.046, +0.046]	+0.010 [−0.042, +0.057]	+0.034
43	0.030	−0.010 [−0.018, −0.003]	0.071	+0.073 [+0.006, +0.176]	+0.084 [+0.014, +0.186]	+0.033
44	0.030	−0.007 [−0.015, +0.000]	0.071	+0.065 [+0.001, +0.168]	+0.072 [+0.008, +0.187]	+0.021

E.4 TRACE

Length groups are defined by the number of valid time steps, which also reflects the recorded span and sampling rate, so they do not isolate duration. Length-group values are point estimates. Pooled values (Table E.9) mix groups that respond differently and are given only for the pooled statements in Section 4.3; each description belongs to a different series, so resampling descriptions also resamples series. The Wilcoxon test is two-sided and Holm-adjusted across the four conditions within each draw; full-shuffle p-values underflow double precision and are reported as $< 10^{-300}$ (the largest adjusted p in the table is 1.7×10^{-138}). Chance is 0.0041.

Table E.10 splits the 168- and 180-point series by recorded time span. Spans are known for 1,203 of 2,006 NOAA test descriptions; all 180-point series with a known span cover six months. The one-week group has 363 order-dependent descriptions; the 364th one-week description is the single unclear one. Table E.11 gives between-group differences in the share lost (intervals resample descriptions within each group); $+0.120$ corresponds to 12 percentage points.

Table E.8: **D2 for TRACE by length group**, per draw: original MRR, share of the original MRR lost, and MRR after full shuffling.

Group	Draw	n	Orig.	Full	MRR after	First-half	Half exch.	Masking
168-point	13	1,050	0.582	0.984	0.009	0.953	0.381	0.691
	14	1,050	0.579	0.982	0.010	0.955	0.385	0.702
	15	1,050	0.580	0.982	0.010	0.965	0.367	0.700
180-point	13	544	0.572	0.972	0.016	0.901	0.819	0.594
	14	544	0.554	0.971	0.016	0.898	0.814	0.598
	15	544	0.551	0.976	0.013	0.886	0.822	0.576
Other lengths	13	411	0.506	0.976	0.012	0.875	0.739	0.604
	14	411	0.478	0.970	0.014	0.899	0.703	0.587
	15	411	0.490	0.962	0.019	0.874	0.733	0.559

Table E.9: **D2 for TRACE on all 2,005 order-dependent descriptions**, per draw.

Condition	Draw	MRR [95% CI]	Share lost [95% CI]	Wilcoxon p (Holm)
Original	13	0.564 [0.546, 0.582]	—	—
	14	0.552 [0.534, 0.570]	—	—
	15	0.554 [0.535, 0.571]	—	—
Full shuffle	13	0.012 [0.009, 0.015]	0.979 [0.974, 0.984]	$< 10^{-300}$
	14	0.013 [0.010, 0.016]	0.977 [0.971, 0.982]	$< 10^{-300}$
	15	0.013 [0.010, 0.016]	0.977 [0.971, 0.982]	$< 10^{-300}$
First-half shuffle	13	0.043 [0.037, 0.049]	0.924 [0.913, 0.935]	8.9×10^{-306}
	14	0.039 [0.033, 0.045]	0.930 [0.918, 0.940]	9.2×10^{-309}
	15	0.040 [0.034, 0.047]	0.927 [0.916, 0.938]	3.3×10^{-304}
Half exchange	13	0.244 [0.229, 0.260]	0.568 [0.539, 0.593]	9.1×10^{-142}
	14	0.243 [0.227, 0.258]	0.559 [0.532, 0.587]	1.7×10^{-138}
	15	0.246 [0.231, 0.261]	0.557 [0.528, 0.584]	2.1×10^{-140}
Masking	13	0.198 [0.184, 0.212]	0.648 [0.625, 0.673]	3.2×10^{-226}
	14	0.191 [0.178, 0.206]	0.653 [0.629, 0.676]	2.0×10^{-230}
	15	0.199 [0.185, 0.212]	0.641 [0.617, 0.665]	2.4×10^{-221}

Table E.10: **D2 for TRACE by recorded time span**, per draw: original MRR and share of the original MRR lost.

Span	Draw	n	Orig.	Full	First-half	Half exch.	Masking
One week (168-pt)	13	363	0.453	0.986	0.957	0.455	0.808
	14	363	0.429	0.980	0.974	0.483	0.807
	15	363	0.453	0.979	0.973	0.475	0.822
28 days (168-pt)	13	323	0.716	0.982	0.956	0.332	0.592
	14	323	0.743	0.985	0.951	0.359	0.664
	15	323	0.718	0.978	0.961	0.300	0.638
Six months (180-pt)	13	278	0.616	0.973	0.887	0.845	0.588
	14	278	0.587	0.978	0.894	0.828	0.639
	15	278	0.585	0.981	0.901	0.854	0.589

Table E.11: **Differences in the share lost between TRACE groups**, per draw (95% intervals).

Comparison	Draw	Full shuffle	First-half	Half exchange	Masking
168-pt − 180-pt	13	+0.012 [−0.001, +0.027]	+0.052 [+0.026, +0.079]	−0.438 [−0.487, −0.383]	+0.098 [+0.040, +0.154]
	14	+0.011 [−0.003, +0.027]	+0.057 [+0.031, +0.087]	−0.429 [−0.480, −0.378]	+0.104 [+0.049, +0.162]
	15	+0.006 [−0.004, +0.019]	+0.079 [+0.054, +0.109]	−0.455 [−0.509, −0.400]	+0.124 [+0.068, +0.182]
1 week − 28 days	13	+0.004 [−0.007, +0.016]	+0.001 [−0.028, +0.028]	+0.123 [+0.025, +0.220]	+0.217 [+0.143, +0.293]
	14	−0.006 [−0.025, +0.009]	+0.023 [−0.003, +0.048]	+0.124 [+0.025, +0.216]	+0.144 [+0.075, +0.216]
	15	+0.001 [−0.018, +0.020]	+0.013 [−0.008, +0.034]	+0.175 [+0.063, +0.280]	+0.184 [+0.110, +0.256]

E.5 ChatTS

ChatTS was run once; accuracy averages the two answer orders, so chance is 0.5. The full-shuffle DiD is in Table 4.5; unlike for CLaSP and TE-3L, its interval resamples signals jointly for the two groups.

Table E.12: **D2 for ChatTS** (one run): accuracy and change from the original condition (95% intervals).

Group (descriptions / series)	Condition	Accuracy	Change
SUSHI, dependent (135/135)	original	0.7333 [0.6704, 0.7963]	—
	full shuffle	0.4667 [0.3963, 0.5370]	−0.2667 [−0.3481, −0.1815]
	first-half shuffle	0.6222 [0.5518, 0.6926]	−0.1111 [−0.1704, −0.0519]
	half exchange	0.6259 [0.5556, 0.6926]	−0.1074 [−0.1556, −0.0556]
	masking	0.6148 [0.5519, 0.6852]	−0.1185 [−0.1815, −0.0556]
	within-patch shuffle	0.7148 [0.6481, 0.7815]	−0.0185 [−0.0556, +0.0185]
	patch-order shuffle	0.5148 [0.4444, 0.5852]	−0.2185 [−0.2963, −0.1406]
SUSHI, invariant (4/4)	original	1.0000 [1.0000, 1.0000]	—
	full shuffle	1.0000 [1.0000, 1.0000]	0.0000 [0.0000, 0.0000]
	first-half shuffle	0.8750 [0.6250, 1.0000]	−0.1250 [−0.3750, 0.0000]
	half exchange	0.7500 [0.5000, 1.0000]	−0.2500 [−0.5000, 0.0000]
	masking	0.8750 [0.6250, 1.0000]	−0.1250 [−0.3750, 0.0000]
	within-patch shuffle	1.0000 [1.0000, 1.0000]	0.0000 [0.0000, 0.0000]
	patch-order shuffle	0.8750 [0.6250, 1.0000]	−0.1250 [−0.3750, 0.0000]
TRUCE, dependent (715/245)	original	0.6238 [0.5932, 0.6554]	—
	full shuffle	0.5112 [0.4813, 0.5407]	−0.1126 [−0.1489, −0.0754]
	first-half shuffle	0.5874 [0.5556, 0.6157]	−0.0364 [−0.0625, −0.0099]
	half exchange	0.4497 [0.4168, 0.4819]	−0.1741 [−0.2151, −0.1354]
	masking	0.6056 [0.5737, 0.6366]	−0.0182 [−0.0377, +0.0007]
TRUCE, invariant (18/14)	original	0.5278 [0.3333, 0.7222]	—
	full shuffle	0.7500 [0.5294, 0.9286]	+0.2222 [+0.0526, +0.4287]
	first-half shuffle	0.6667 [0.5000, 0.8333]	+0.1389 [0.0000, +0.2941]
	half exchange	0.5833 [0.3529, 0.7943]	+0.0556 [−0.1579, +0.2812]
	masking	0.5556 [0.3750, 0.7188]	+0.0278 [−0.0882, +0.1579]

Table E.13: **Equivalence tests for ChatTS in D2**, including the prompt-number control (series unchanged; only the removed mean and scale in the prompt shifted by the amount masking causes; 100 descriptions per dataset, over 100 SUSHI and 88 TRUCE series).

Data	Comparison	Desc./series	Change	90% CI	Outcome
SUSHI	within-patch shuffle (dependent)	135/135	−0.0185	[−0.0481, +0.0111]	equivalent
TRUCE	masking (dependent)	715/245	−0.0182	[−0.0345, −0.0027]	equivalent
SUSHI	prompt numbers shifted	100/100	−0.0050	[−0.0150, 0.0000]	equivalent
TRUCE	prompt numbers shifted	100/88	0.0000	[−0.0248, +0.0211]	equivalent

Appendix F

Supplementary Results for D3

The CLaSP ladder uses all queries; equivalence tests use order-dependent queries (135 SUSHI, 715 TRUCE). The share lost is $1 - P_c/P_{\text{orig}}$. Unless stated otherwise, intervals are 95% percentile bootstrap intervals over signals, and equivalence tests use 90% intervals and the ± 0.05 margin.

F.1 CLaSP

Table F.1: **D3 ladder for CLaSP**, all queries, per checkpoint (MRR; 10,000 resamples). Full-shuffle values include all queries and therefore differ slightly from Table E.2.

Data	Ckpt	Queries/series	Original	Full shuffle	Own-value resample	Matched Gaussian
SUSHI	42	140/140	0.304	0.083 [0.052, 0.119]	0.078 [0.050, 0.110]	0.016 [0.010, 0.024]
	43		0.311	0.076 [0.049, 0.107]	0.069 [0.042, 0.099]	0.017 [0.008, 0.033]
	44		0.368	0.093 [0.063, 0.126]	0.121 [0.082, 0.165]	0.026 [0.014, 0.044]
TRUCE	42	738/246	0.099	0.034 [0.024, 0.046]	0.029 [0.022, 0.037]	0.024 [0.018, 0.031]
	43		0.105	0.032 [0.025, 0.039]	0.031 [0.025, 0.038]	0.023 [0.017, 0.031]
	44		0.107	0.028 [0.022, 0.034]	0.023 [0.018, 0.028]	0.025 [0.019, 0.032]

Table F.2: **Gaussian condition against the reference levels, and resample − Gaussian**, CLaSP, all queries. Chance 0.0169; length-only reference 0.0394 (SUSHI), 0.0247 (TRUCE).

Data	Ckpt	Matched Gaussian	Chance inside	Len. ref. inside	Resample − Gaussian
SUSHI	42	0.016 [0.010, 0.024]	yes	no	+0.062 [+0.033, +0.096]
	43	0.017 [0.008, 0.033]	yes	no	+0.052 [+0.020, +0.085]
	44	0.026 [0.014, 0.044]	yes	yes	+0.094 [+0.052, +0.141]
TRUCE	42	0.024 [0.018, 0.031]	no	yes	+0.005 [−0.005, +0.015]
	43	0.023 [0.017, 0.031]	no	yes	+0.008 [−0.002, +0.017]
	44	0.025 [0.019, 0.032]	no	yes	−0.002 [−0.011, +0.006]

Table F.4 compares SUSHI signals with spikes (negative, positive or both; 60 signals) with clean or smooth signals (39; the constant signal excluded). Retention is the original rank divided by the rank under resampling for each signal's single query (1 = no loss). The difference interval resamples signals within each group (10,000 resamples); the Mann–Whitney test is one-sided (spike signals retain more). No interval excludes zero.

Table F.3: **Equivalence tests for CLaSP**: share lost under resampling minus share lost under full shuffling (order-dependent queries). Positive: resampling removes more.

Data	Ckpt	Difference	90% CI	Outcome
SUSHI	42	+0.0080	[−0.0592, +0.0783]	inconclusive
	43	+0.0272	[−0.0300, +0.0939]	inconclusive
	44	−0.0827	[−0.1501, −0.0239]	shifted
TRUCE	42	+0.0614	[−0.0457, +0.1719]	inconclusive
	43	+0.0115	[−0.0594, +0.0901]	inconclusive
	44	+0.0579	[+0.0023, +0.1177]	shifted

Table F.4: **Retention under own-value resampling for SUSHI signals with and without spikes**, CLaSP.

Ckpt	Spike mean	Clean mean	Difference [95% CI]	Spike median	Clean median	Mann–Whitney p
42	0.375	0.522	−0.147 [−0.513, +0.186]	0.032	0.029	0.234
43	0.321	0.299	+0.023 [−0.201, +0.254]	0.032	0.016	0.061
44	0.821	0.606	+0.214 [−0.408, +0.961]	0.066	0.034	0.142

F.2 TE-3L negative control

Table F.5: **Equivalence tests for TE-3L in D3** (order-dependent queries; all 12 equivalent; 2,000 resamples). Original MRR: 0.0037 (SUSHI), 0.0295 (TRUCE). Largest absolute change 0.0067.

Data	Condition	Draw 42: change [90% CI]	Draw 43	Draw 44
SUSHI	own-value resample	−0.0004 [−0.0007, −0.0002]	−0.0005 [−0.0008, −0.0002]	−0.0005 [−0.0008, −0.0002]
	matched Gaussian	−0.0000 [−0.0004, +0.0004]	−0.0002 [−0.0005, +0.0000]	−0.0004 [−0.0007, −0.0001]
TRUCE	own-value resample	−0.0067 [−0.0142, +0.0009]	−0.0033 [−0.0126, +0.0061]	−0.0040 [−0.0122, +0.0043]
	matched Gaussian	−0.0026 [−0.0123, +0.0082]	−0.0028 [−0.0123, +0.0072]	−0.0055 [−0.0139, +0.0028]

F.3 TRACE

Table F.6 gives point estimates by length group. The last column places the Gaussian MRR on a scale from chance (0) to the length-only reference (1); negative values lie below chance. Chance is 0.0041; length-only references are 0.0072 (168-point, 1,051 candidates) and 0.0126 (180-point, 544 candidates). Full-shuffle values are those of Table E.8.

Table F.6: **D3 ladder for TRACE by length group**, per draw (MRR, order-dependent queries).

Group	Draw	n	Original	Full shuffle	Resample	Gaussian	Gaussian position
168-point	13	1,050	0.5821	0.0094	0.0083	0.0039	−0.05
	14	1,050	0.5791	0.0103	0.0068	0.0035	−0.20
	15	1,050	0.5800	0.0103	0.0124	0.0060	+0.63
180-point	13	544	0.5720	0.0162	0.0119	0.0038	−0.03
	14	544	0.5537	0.0161	0.0161	0.0044	+0.04
	15	544	0.5506	0.0133	0.0115	0.0030	−0.12
Other lengths	13	411	0.5058	0.0122	0.0131	0.0106	—
	14	411	0.4782	0.0143	0.0127	0.0134	—
	15	411	0.4904	0.0187	0.0137	0.0079	—

Table F.7: **D3 ladder for TRACE**, all 2,005 order-dependent queries, per draw (2,000 resamples).

Condition	Draw	MRR [95% CI]	Share lost [95% CI]
Original	13	0.5637 [0.5464, 0.5823]	—
	14	0.5515 [0.5330, 0.5694]	—
	15	0.5537 [0.5356, 0.5717]	—
Full shuffle	13	0.0118 [0.0093, 0.0147]	0.979 [0.974, 0.984]
	14	0.0127 [0.0097, 0.0161]	0.977 [0.971, 0.982]
	15	0.0128 [0.0098, 0.0162]	0.977 [0.970, 0.982]
Own-value resample	13	0.0103 [0.0083, 0.0126]	0.982 [0.978, 0.985]
	14	0.0105 [0.0081, 0.0134]	0.981 [0.976, 0.985]
	15	0.0124 [0.0096, 0.0155]	0.978 [0.972, 0.983]
Matched Gaussian	13	0.0052 [0.0039, 0.0070]	0.991 [0.987, 0.993]
	14	0.0058 [0.0043, 0.0076]	0.990 [0.986, 0.992]
	15	0.0056 [0.0040, 0.0076]	0.990 [0.986, 0.993]

Table F.8: **Paired tests between ladder steps for TRACE** (Wilcoxon signed-rank, two-sided, Holm-adjusted within draw; 2,005 order-dependent queries). In draw 14, ranks under resampling and shuffling differ ($p = 0.044$), but the difference in share lost is within the equivalence margin (Table F.9).

Draw	Resample vs. full shuffle	Resample vs. Gaussian	Full shuffle vs. Gaussian
13	0.613	2.3×10^{-25}	2.9×10^{-24}
14	0.044	3.4×10^{-23}	1.9×10^{-30}
15	0.993	9.5×10^{-25}	1.3×10^{-25}

Table F.9: **Equivalence tests for TRACE**: share lost under resampling minus under full shuffling (2,005 order-dependent queries).

Draw	Difference	90% CI	Outcome
13	+0.0027	[−0.0019, +0.0075]	equivalent
14	+0.0039	[−0.0015, +0.0098]	equivalent
15	+0.0008	[−0.0055, +0.0069]	equivalent

Table F.10: **Differences in the share lost between TRACE's 168- and 180-point groups** (168 − 180), per draw. Full-shuffle intervals come from a separate bootstrap and differ slightly from Table E.11; point estimates are identical.

Draw	Full shuffle	Own-value resample	Matched Gaussian
13	+0.012 [−0.001, +0.026]	+0.006 [−0.002, +0.016]	−0.000 [−0.003, +0.003]
14	+0.011 [−0.003, +0.028]	+0.017 [+0.005, +0.033]	+0.002 [−0.001, +0.007]
15	+0.006 [−0.005, +0.018]	−0.000 [−0.012, +0.012]	−0.005 [−0.011, −0.000]

F.4 ChatTS

Accuracy over 140 SUSHI queries (140 series) and 738 TRUCE queries (246 series), with 2,000 resamples over series. Original and full-shuffle values use all queries and so differ slightly from Table E.12. Conditions A, B and C are defined in Section C.4.

Table F.11: **D3 for ChatTS** (one run): accuracy (95% intervals) and whether the interval includes chance.

Data	Condition	Accuracy	Chance inside
SUSHI	original	0.7429 [0.6821, 0.8071]	no
	full shuffle	0.4857 [0.4179, 0.5571]	yes
	own-value resample	0.4714 [0.4036, 0.5394]	yes
	matched Gaussian	0.4643 [0.3929, 0.5357]	yes
	five-field summary (text only)	0.4821 [0.4214, 0.5464]	yes
	A: Gaussian series, original prompt numbers	0.4679 [0.3964, 0.5357]	yes
	B: original series, another series' numbers	0.7393 [0.6786, 0.8000]	no
	C: Gaussian series, its own numbers	0.4643 [0.3929, 0.5286]	yes
TRUCE	original	0.6220 [0.5901, 0.6531]	no
	full shuffle	0.5163 [0.4885, 0.5454]	yes
	own-value resample	0.5115 [0.4810, 0.5413]	yes
	matched Gaussian	0.5007 [0.4688, 0.5332]	yes
	five-field summary (text only)	0.5190 [0.4919, 0.5461]	yes
	A: Gaussian series, original prompt numbers	0.4973 [0.4641, 0.5285]	yes
	B: original series, another series' numbers	0.6159 [0.5854, 0.6484]	no
	C: Gaussian series, its own numbers	0.5007 [0.4695, 0.5325]	yes

Table F.12: **Paired comparisons for ChatTS in D3** (difference in accuracy; agreement = share of questions answered with the same letter). C reproduces the standard Gaussian exactly; A ≈ C; B is equivalent to the original; the five-field summary falls well below the original.

Data	Comparison	Difference	95% CI	90% CI	Outcome	Agreement
SUSHI	C − standard Gaussian	0.0000	[0.0000, 0.0000]	[0.0000, 0.0000]	equivalent	1.000
	A − C	+0.0036	[−0.0107, +0.0214]	[−0.0071, +0.0179]	equivalent	0.982
	B − original	−0.0036	[−0.0286, +0.0179]	[−0.0216, +0.0143]	equivalent	—
	summary − original	−0.2607	[−0.3429, −0.1857]	[−0.3250, −0.2036]	shifted	—
TRUCE	C − standard Gaussian	0.0000	[0.0000, 0.0000]	[0.0000, 0.0000]	equivalent	1.000
	A − C	−0.0034	[−0.0122, +0.0054]	[−0.0102, +0.0034]	equivalent	0.974
	B − original	−0.0061	[−0.0169, +0.0047]	[−0.0149, +0.0027]	equivalent	—
	summary − original	−0.1030	[−0.1423, −0.0664]	[−0.1342, −0.0718]	shifted	—

Appendix G

Pre-registered Predictions and Their Outcomes

Table G.1 lists the 25 predictions about diagnostic results recorded before the analyses that tested them: 15 were confirmed and 10 missed. Identifiers match the repository, where confirmed and missed appear as `CONFIRMED/HIT` and `MISSED/MISS`; P5 comes from the TRACE D1 item set. Expectations about the validation process (e.g. screening pass rates) and the execution of single commands (counts, file sizes) are excluded, with one exception kept as recorded: the ChatTS inference-time estimate (E-time), which was recorded together with the ChatTS predictions. No prediction was recorded about which components would fail for ChatTS in D1.

Table G.1: **Predictions about diagnostic results**, recorded before the analyses (15 confirmed, 10 missed).

ID	Diagnostic, model	Prediction	Result	Outcome	If missed: what was found
P5	D1, TRACE on NOAA	Location-swap (N5) accuracy lies in $[0.45, 0.60]$, near chance.	0.925, 0.935, 0.918	missed	Location swaps are discriminable under the tested setup (Section 4.2.1).
PC1-1	D1, ChatTS on SUSHI	Random-control accuracy ≥ 0.90.	0.726 $[0.695, 0.755]$	missed	Capability present, far above chance, but lower than expected.
P2-1	D2, CLaSP on SUSHI	Full shuffle: order-dependent group loses > 0.05 of its MRR at every checkpoint.	Share lost 0.764–0.783; change -0.280 to -0.232	confirmed	—
P2-2	D2, CLaSP on TRUCE	Full shuffle: $\mathrm{DiD} > 0$ at every checkpoint.	$+0.002$, $+0.076$, $+0.073$	confirmed	—
P2-3	D2, CLaSP	Full shuffle removes at least as much MRR as first-half shuffle, both datasets, every checkpoint.	6 of 6 (Table E.2)	confirmed	—
P2-4	D2, TE-3L	All 24 dependent comparisons equivalent; no change reaches 0.05.	24/24; largest 0.0105	confirmed	—

Table G.1 continued

ID	Diagnostic, model	Prediction	Result	Outcome	If missed: what was found
P2-5	D2, CLaSP on TRUCE	Full shuffle: invariant group equivalent to no change at every checkpoint.	Equivalent at 43, 44; not at 42 (90% CI $[-0.180, -0.005]$)	missed	At checkpoint 42, most of the invariant loss comes from one description (Section 4.3).
P2-6	D2, SUSHI grouping	At most 14 order-invariant test descriptions.	4	confirmed	—
P2-7	D2, NOAA grouping	$< 5\%$ of test descriptions order-invariant.	0 of 2,006	confirmed	—
P2-8	D2, TRUCE grouping	$\geq 15\%$ of classified descriptions order-invariant.	3.3% (245 of 7,334)	missed	Order-invariant descriptions are rare in all datasets.
P2-9	D2, TRACE on NOAA	Full shuffle: the 2,005 dependent descriptions lose > 0.05 of MRR in every draw.	Share lost 0.977–0.979	confirmed	—
PJ	D2, ChatTS	Shifting only the prompt numbers by the masking amount is equivalent to no change on both datasets.	SUSHI -0.005, TRUCE 0.000; both equivalent	confirmed	—
P3-1	D3, CLaSP	Matched Gaussian falls to chance: 95% interval of its ratio to chance includes 1 at every checkpoint.	Only at checkpoint 43 (1.05–1.68; 0.99–1.70; 1.17–1.86 × chance)	missed	The Gaussian keeps length; on TRUCE it reaches the length-only reference, then treated as a length-only condition (Section 3.8).
P3-2a	D3, TRACE	Resampling and full shuffle remove equivalent shares in every draw.	Equivalent in all 3	confirmed	—
P3-2b	D3, CLaSP on SUSHI	Same, order-dependent queries, every checkpoint.	42, 43: inconclusive; 44: shifted	missed	Intervals too wide at two checkpoints; at 44 resampling removes less (Table F.3).
P3-2c	D3, CLaSP on TRUCE	Same, order-dependent queries, every checkpoint.	42, 43: inconclusive; 44: shifted	missed	Resampling removes more at all checkpoints, clearly only at 44; may reflect 12-value resampling (untested).
P3-3	D3, TRACE	Under resampling, MRR is 2.0–3.5× chance in every draw.	2.52, 2.58, 3.04 ×; all above chance	confirmed	—
P3-4	D3, CLaSP	Under resampling, lower bound of MRR/chance > 1 at every checkpoint.	1.72, 1.79, 1.81	confirmed	—
P3-5	D3, CLaSP on SUSHI	Spike signals retain more rank than clean/smooth ones at every checkpoint.	Higher at 43, 44; lower at 42; no interval excludes 0	missed	No clear difference (Table F.4).

Table G.1 continued

ID	Diagnostic, model	Prediction	Result	Outcome	If missed: what was found
P3-6	D3, TRACE	Resampling share lost does not differ between 168- and 180-point groups (interval includes 0 in every draw).	Includes 0 in draws 13, 15; excludes in 14 ($+0.017$)	missed	Small difference in draw 14 (Table F.10).
P3-7	D3, TE-3L	All 12 dependent comparisons equivalent; no change reaches 0.05.	12/12; largest 0.0067	confirmed	—
A≈C	D3, ChatTS	Conditions A and C equivalent on both datasets, since a matched Gaussian has nearly the original prompt numbers.	SUSHI $+0.004$, TRUCE -0.003; equivalent	confirmed	—
PC1-2	ChatTS, position	Share of answers choosing A lies in 0.50 ± 0.15.	0.442	confirmed	—
PC1-3	ChatTS, readout	Logit readout agrees with generated answer on $\geq$ 95% of checked questions per diagnostic.	200/200 in each	confirmed	—
E-time	ChatTS, execution	All 31,356 questions take 1–3 GPU hours.	26.4 min	missed	Estimate too high; an execution estimate, not a result prediction.

Table G.2 adds three expectations recorded for checks within an analysis rather than as predictions of a diagnostic's outcome; they are not counted above.

Table G.2: **Expectations for checks within an analysis** (not counted in Table G.1).

ID	Expectation	Result	Outcome	Note
S2	D2, TRACE: in both length groups, losses order as full > first-half > masking > half exchange, every draw.	Holds for 168-point; for 180-point, half exchange > masking in every draw	missed	The order of half exchange and masking reverses between groups (Section 4.3).
D2 (span)	D2, TRACE: within 168-point series, half-exchange share lost differs between one-week and 28-day series by < 15 points (mean over draws).	Mean 14.0 (12.3, 12.4, 17.5)	confirmed	Borderline: draw 15 alone exceeds 15.
P-R1	D1, ChatTS: certified C4 accuracy within ± 0.03 of the all-items 0.658.	0.690 ($+0.032$)	missed	Excluded items scored lower (census failures 0.636, generic 0.551, cross-slot 0.175).

Appendix H

Code, Data and Reproducibility

Code and result files are in a public repository, https://github.com/RaneeLuo/Thesis-probes. It contains the code that trains the CLaSP reimplementation, constructs the diagnostic items and conditions, runs each model and performs the analyses. Result files are in three directories: `results/experiments/` (per-run records, summaries and statistics), `results/analysis/` (item construction, human validation, description grouping, supplementary analyses) and `results/logs/` (ChatTS GPU logs). The README points to the command that reproduces each step and to the package versions used. In the repository, D1–D3 are named `probe1`–`probe3`, and the five-field summary is named `five_number`. All random draws are seeded per series, condition and run (Chapter C). Answering all 31,356 ChatTS questions took 26.4 GPU minutes.

Table H.1: **Model versions.**

Model	Version used
CLaSP	Reimplementation in this repository (`models/clasp/`), trained with seeds 42, 43 and 44; each run keeps the checkpoint with the lowest validation loss, referred to by its seed.
TRACE	Checkpoint released by the authors, `results/model_checkpoints/context_align/retriever_demo.pt` (46,271,904 bytes), in https://github.com/Graph-and-Geometric-Learning/TRACE-Multimodal-TSEncoder, with the text encoder recorded in the checkpoint, `nomic-ai/nomic-embed-text-v1.5`.
ChatTS-14B	`bytedance-research/ChatTS-14B` on Hugging Face, revision `1e661101dcfff86dc66f3397336b85f2f1cc5e89` (24 July 2025).
`text-embedding-3-large`	OpenAI embeddings API, default output size, queried between 30 July and 15 August 2026; results refer to the model served under this name at that time.